

Ellipsoidal cone eigenvalue complementarity problems: Solvability conditions, reformulations and numerical experiments

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ABSTRACT

This paper investigates two variants of the ellipsoidal cone eigenvalue complementarity problem (ECEiCP). We first derive sufficient conditions for solution existence and then establish equivalent reformulations by leveraging connections between ellipsoidal cones and second-order cones, which can be reformulated as nonlinear second-order conic complementarity problems (NSOC-CPs). Numerical experiments on synthetic instances show that these counterpart complementarity problems can be solved efficiently by a customized projection method with an exact residual-based stopping rule.

1. Introduction

This paper studies the following ellipsoidal cone eigenvalue complementarity problems (ECEiCPs)

$$x \in \mathcal{K}_{\mathcal{E}}, Ax - \lambda x \in \mathcal{K}_{\mathcal{E}}^*, \langle x, Ax - \lambda x \rangle = 0, \quad (1)$$

$$x \in \mathcal{K}_{\mathcal{E}}, \lambda^2 Ax + \lambda Bx + Cx \in \mathcal{K}_{\mathcal{E}}^*, \langle x, \lambda^2 Ax + \lambda Bx + Cx \rangle = 0, \quad (2)$$

where $A, B, C \in \mathbb{R}^{n \times n}$ are given matrices, and $\mathcal{K}_{\mathcal{E}}$ denotes an n -dimensional ellipsoidal cone defined by

$$\mathcal{K}_{\mathcal{E}} := \{x \in \mathbb{R}^n : \langle x, Qx \rangle \leq 0, \langle u_n, x \rangle \geq 0\} \quad (3)$$

with $Q \in \mathbb{R}^{n \times n}$ being a real-valued nonsingular symmetric matrix having $n - 1$ positive eigenvalues and exactly one negative eigenvalue corresponding to the eigenvector $u_n \in \mathbb{R}^n$. The inner product $\langle \cdot, \cdot \rangle$ is the standard Euclidean inner product, and the dual cone $\mathcal{K}_{\mathcal{E}}^*$ is given by

$$\mathcal{K}_{\mathcal{E}}^* := \{y \in \mathbb{R}^n : \langle x, y \rangle \geq 0 \text{ for all } x \in \mathcal{K}_{\mathcal{E}}\}.$$

Our investigation is motivated by three principal considerations: (a) Ellipsoidal cones, which are nonsymmetric under the standard Euclidean inner product [1], generalize several well-known cones, such as second-order cones [2–7], circular cones [8–14], and elliptic cones [15] through parameter variations in Q and u_n (see Table 1 in Section 2). Consequently, ellipsoidal cone optimization provides a unifying framework for theoretical analysis [1,16–18] and applications [19–21] across these special cases; (b) Eigenvalue

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complementarity problems constitute a significant branch of eigenvalue theory [22] that integrates optimization principles with complementarity constraints. This formalism yields rigorous characterizations for diverse problems with substantial applications in engineering and economics [23–26], including dynamic analysis of structural mechanical systems, vibro-acoustic systems, electrical circuit simulation, fluid dynamics, and mechanical contact problems; (c) Cone eigenvalue complementarity problems have been a prominent research focus, with significant progress made in theoretical analysis, algorithm design, and convergence analysis, particularly for Pareto and second-order cone variants; see [27–33] and references therein. However, existing research has focused mainly on symmetric cones in $\mathbb{R}^n$, leaving nonsymmetric conic extensions largely unexplored. Addressing this gap is a primary objective of our work.

In this paper, we investigate two variants of the ellipsoidal cone eigenvalue complementarity problem. Our contributions include the following three parts:

1. Solvability conditions (see Theorem 1):
 - For any given $A \in \mathbb{R}^{n \times n}$, problem (1) has solutions.
 - If A is $\mathcal{K}_\mathcal{E}$ -regular and the triple (A, B, C) is $\mathcal{K}_\mathcal{E}$ -hyperbolic, problem (2) has solutions.
 - 0 is an eigenvalue of problem (2) iff $C \in R'_0(\mathcal{K}_\mathcal{E})$.
 - If $C \in S'_0(\mathcal{K}_\mathcal{E})$ and A is strictly $\mathcal{K}_\mathcal{E}$ -copositive, then there exists at least one positive and one negative eigenvalue of problem (2).
 - If $A \in S'_0(\mathcal{K}_\mathcal{E})$ and C is strictly $\mathcal{K}_\mathcal{E}$ -copositive, then there exists at least one positive and one negative eigenvalue of problem (2).
2. Model reformulations (see Theorems 2 and 3): To exclude the trivial solution (the null vector), their regularized versions are proposed (models (10) and (11)), which can be reformulated as nonlinear second-order conic complementarity problems by leveraging connections between ellipsoidal cones and second-order cones.
3. Numerical experiments (see Section 4): To show the solvability of these ellipsoidal cone eigenvalue complementarity problems, we propose a customized projection method for the associated NSOCCPs; for problem (2) the numerical solution is obtained from the equivalent lifted linear form of Lemma 4 over the product of second-order cones. Numerical experiments are conducted to validate the effectiveness of the proposed algorithm.

Although there exists a linear equivalence that allows transformation of an ellipsoidal cone into a second-order cone (see Lemma 1), the resulting NSOCCP involves a nonlinear mapping that is not simply a change of variables of a known second-order cone eigenvalue problem. Moreover, the nonsymmetry of ellipsoidal cone leads to a non-self-dual transformed structure and the need for careful treatment of the dual cone in the reformulation. The corresponding solvability conditions are derived directly for the ellipsoidal geometry, and the subsequent algorithmic analysis addresses the specific non-differentiability at the origin and the discriminant singularities in the quadratic case (see Theorem 1, Remark 3 and Lemma 8). Therefore, the results are not direct corollaries of existing second-order cone theory.

The paper is structured as follows. Section 2 presents preliminary results on ellipsoidal cones and derives Jacobian (or gradient) chain rules for a specialized class of composite functions. Our main contributions are presented in Section 3 including the solvability conditions and reformulated models. Section 4 shows the performance of our proposed method. Concluding remarks appear in Section 5.

2. Preliminaries

This section reviews essential background on ellipsoidal cones and presents a key technical lemma for subsequent analysis. Additional details appear in [1].

From the definition of the ellipsoidal cone in Section 1, its inner and boundary are

$$\begin{aligned} \text{int}\mathcal{K}_\mathcal{E} &:= \{x \in \mathbb{R}^n : x^T Q x < 0, u_n^T x > 0\}, \\ \text{bd}\mathcal{K}_\mathcal{E} &:= \{x \in \mathbb{R}^n : x^T Q x = 0, u_n^T x > 0\} \cup \{0\}. \end{aligned} \quad (4)$$

Consider the symmetric matrix $Q \in \mathbb{R}^{n \times n}$ with the orthogonal decomposition

$$Q = \sum_{i=1}^n \lambda_i u_i u_i^T,$$

where the eigenpairs (λ_i, u_i) satisfy both the spectral ordering

$$\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_{n-1} > 0 > \lambda_n \quad (5)$$

and the orthogonality condition

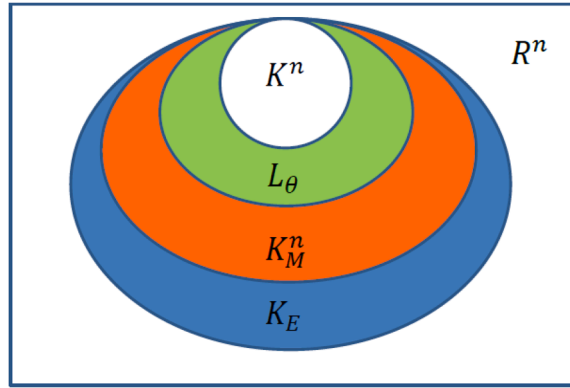
$$u_i^T u_j = \delta_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j. \end{cases} \quad (6)$$

As a fundamental closed convex cone, the ellipsoidal cone $\mathcal{K}_\mathcal{E}$ generalizes several important cones including second-order cones, circular cones, and elliptic cones (Table 1). These satisfy the inclusion relations

$$\mathcal{K}^n \subseteq \mathcal{L}_\theta \subseteq \mathcal{K}_M^n \subseteq \mathcal{K}_\mathcal{E} \subseteq \mathbb{R}^n,$$

Table 1The expression of three convex cones and the associated parameters Q, u_n in (3).

Name	Expression	Q	u_n
Second-order cone	$\mathcal{K}^n := \{(\bar{x}, x_n) \in \mathbb{R}^{n-1} \times \mathbb{R} : \ \bar{x}\ \leq x_n\}$	$\begin{bmatrix} I_{n-1} & 0 \\ 0 & -1 \end{bmatrix}$	$u_n = e_n$
Circular cone	$\mathcal{L}_\theta := \{(\bar{x}, x_n) \in \mathbb{R}^{n-1} \times \mathbb{R} : \ \bar{x}\ \leq x_n \tan \theta\}$	$\begin{bmatrix} I_{n-1} & 0 \\ 0 & -\tan^2 \theta \end{bmatrix}$	$u_n = e_n$
Elliptic cone	$\mathcal{K}_M^n := \{(\bar{x}, x_n) \in \mathbb{R}^{n-1} \times \mathbb{R} : \ M\bar{x}\ \leq x_n\}$	$\begin{bmatrix} M^T M & 0 \\ 0 & -1 \end{bmatrix}$	$u_n = e_n$

**Fig. 1.** The relations among $\mathcal{K}^n$, $\mathcal{L}_\theta$, $\mathcal{K}_M^n$, and $\mathcal{K}_E$.

where $\|\bar{x}\|$ denotes the Euclidean norm of $\bar{x} \in \mathbb{R}^{n-1}$, $\theta \in (0, \pi/2)$, and $M \in \mathbb{R}^{(n-1) \times (n-1)}$ is nonsingular. Thus, the ellipsoidal cone $\mathcal{K}_E$ provides a unifying framework that naturally extends these conical structures, as illustrated in Fig. 1.

The following lemma characterizes the transformation between ellipsoidal cones and second-order cones, derived from [1, Theorems 2.1 and 2.2].

Lemma 1. Let $\mathcal{K}_E$ and $\mathcal{K}^n$ be the ellipsoidal cone and second-order cone defined as above. Then, we have

- (a) $\mathcal{K}_E = T\mathcal{K}^n$ and $\mathcal{K}^n = T^{-1}\mathcal{K}_E$;
- (b) $\mathcal{K}_E^* = (T^T)^{-1}\mathcal{K}^n$ and $\mathcal{K}^n = T^T\mathcal{K}_E^*$;
- (c) $\mathcal{K}_E^* = \{y \in \mathbb{R}^n : y^T Q^{-1}y \leq 0, u_n^T y \geq 0\}$,

where $T := UD$, $U := [u_1, u_2, \dots, u_n]$, $D := \text{diag}((\lambda_1)^{-1/2}, \dots, (\lambda_{n-1})^{-1/2}, (-\lambda_n)^{-1/2})$ with eigenpairs (λ_i, u_i) of Q satisfying the relations (5) and (6), and $\mathcal{K}_E^*$ denotes the dual cone of $\mathcal{K}_E$.

We conclude this section with a technical lemma on Jacobian (or gradient) calculus for a special type of composition function, whose proof follows from basic calculus and is omitted.

Lemma 2. Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and $g_i : \mathbb{R}^n \rightarrow \mathbb{R}$ ($1 \leq i \leq n$) be real-valued differentiable functions. Define

$$G(x) := \begin{bmatrix} g_1(x) \\ \vdots \\ g_n(x) \end{bmatrix}, \quad H(x) := f(x)G(x) = \begin{bmatrix} f(x)g_1(x) \\ \vdots \\ f(x)g_n(x) \end{bmatrix}.$$

Then H is differentiable on $\mathbb{R}^n$ with the Jacobian mapping and its gradient

$$JH(x) = G(x)(\nabla f(x))^T + f(x)JG(x), \quad \nabla H(x) = [JH(x)]^T,$$

where $JG(x)$ denotes the Jacobian of G .

3. Main results

This section establishes theoretical foundations for ellipsoidal cone eigenvalue complementarity problems, presenting both solvability conditions and model reformulations.

3.1. Solvability conditions

The solvability conditions of problem (1) and problem (2) are considered in this subsection. In order to clearly describe our results, we need a few concepts adapted from [34, Definition 2.1 and Definition 2.5] and [31, Definition 1].

Definition 1. Let $\mathcal{K}_\varepsilon$ be an ellipsoidal cone defined as in (3).

- (a) A matrix $A \in \mathbb{R}^{n \times n}$ is called $\mathcal{K}_\varepsilon$ -regular, if $\langle x, Ax \rangle \neq 0$ for all $x \in \mathcal{K}_\varepsilon \setminus \{0\}$.
- (b) A matrix $A \in \mathbb{R}^{n \times n}$ is called strictly $\mathcal{K}_\varepsilon$ -copositive, if $\langle x, Ax \rangle > 0$ for all $x \in \mathcal{K}_\varepsilon \setminus \{0\}$.
- (c) A triple (A, B, C) with $A, B, C \in \mathbb{R}^{n \times n}$ is called $\mathcal{K}_\varepsilon$ -hyperbolic, if

$$\langle x, Bx \rangle^2 - 4\langle x, Ax \rangle \langle x, Cx \rangle \geq 0$$

for all $x \in \mathcal{K}_\varepsilon \setminus \{0\}$.

- (d) The class $R_0(\mathcal{K}_\varepsilon)$ consists of those matrices $A \in \mathbb{R}^{n \times n}$ such that there exists no nonzero $x \in \mathcal{K}_\varepsilon$ satisfying $Ax \in \mathcal{K}_\varepsilon^*$ and $\langle x, Ax \rangle = 0$.
- (e) The class $R'_0(\mathcal{K}_\varepsilon)$ consists of those matrices $A \in \mathbb{R}^{n \times n}$ such that $\langle x, Ax \rangle = 0$ for at least a nonzero $x \in \mathcal{K}_\varepsilon$ satisfying $Ax \in \mathcal{K}_\varepsilon^*$.
- (f) The class $S_0(\mathcal{K}_\varepsilon)$ consists of those matrices $A \in \mathbb{R}^{n \times n}$ such that $Ax \in \mathcal{K}_\varepsilon^*$ for at least a nonzero $x \in \mathcal{K}_\varepsilon$.
- (g) The class $S'_0(\mathcal{K}_\varepsilon)$ consists of those matrices $A \in \mathbb{R}^{n \times n}$ such that there exists no nonzero $x \in \mathcal{K}_\varepsilon$ satisfying $Ax \in \mathcal{K}_\varepsilon^*$.

Recent literature has addressed solvability conditions for eigenvalue problems over general cones [34–36]. In line with these developments, before establishing the sufficient conditions for the existence of a solution for problem (1) and problem (2), we need the following two technical lemmas, whose proofs can be found in [37, Theorem 3.1.1] and [34, Lemma 2.3], respectively.

Lemma 3. Let K be a compact convex subset of a finite-dimensional space X , and $\varphi : X \times X \mapsto \mathbb{R}$ be a function satisfying

- $$\begin{cases} (a) & \varphi(\cdot, y) \text{ is upper semicontinuous for each fixed } y \in K; \\ (b) & \varphi(x, \cdot) \text{ is convex for each fixed } x \in K; \\ (c) & \varphi(y, y) \geq 0 \text{ for all } y \in K. \end{cases}$$

Then there exists $\bar{x} \in K$ such that $\varphi(\bar{x}, y) \geq 0$ for all $y \in K$.

Lemma 4. Consider problem (2) with $A, B, C \in \mathbb{R}^{n \times n}$. Take any symmetric and positive definite matrix $E \in \mathbb{R}^{n \times n}$ and define matrices $J, G, H \in \mathbb{R}^{2n \times 2n}$ as

$$J := \begin{bmatrix} A & 0 \\ 0 & E \end{bmatrix}, \quad G := \begin{bmatrix} -B & -C \\ E & 0 \end{bmatrix}, \quad H := \begin{bmatrix} B & -C \\ E & 0 \end{bmatrix}.$$

(i) Assume that (λ, x) solves problem (2) and consider J, G, H defined as above.

(a) If $\lambda = 0$, then $(0, 0, x) \in \mathbb{R} \times \mathbb{R}^n \times \mathbb{R}^n$ is a solution of the following problems

$$\begin{cases} y := (u, v) \in \mathcal{K}_\varepsilon \times \mathcal{K}_\varepsilon, \quad \lambda Jy - Gy \in \mathcal{K}_\varepsilon^* \times \mathcal{K}_\varepsilon^*, \\ \langle y, \lambda Jy - Gy \rangle = 0, \quad a^T(u + v) = 1; \end{cases} \quad (7)$$

$$\begin{cases} y := (u, v) \in \mathcal{K}_\varepsilon \times \mathcal{K}_\varepsilon, \quad \lambda Jy - Hy \in \mathcal{K}_\varepsilon^* \times \mathcal{K}_\varepsilon^*, \\ \langle y, \lambda Jy - Hy \rangle = 0, \quad a^T(u + v) = 1. \end{cases} \quad (8)$$

(b) If $\lambda > 0$, then $(\lambda, (1 + \lambda)^{-1}\lambda x, (1 + \lambda)^{-1}x)$ solves problem (7).

(c) If $\lambda < 0$, then $(-\lambda, -(1 - \lambda)^{-1}\lambda x, (1 - \lambda)^{-1}x)$ solves problem (8).

(ii) Consider J, G, H defined as above.

(a) If (λ, u, v) solves problem (7) and $\lambda \neq 0$, then $\lambda > 0$ and $(\lambda, (1 + \lambda)v)$ solves problem (2).

(b) If (λ, u, v) solves problem (8) and $\lambda \neq 0$, then $\lambda > 0$ and $(-\lambda, (1 + \lambda)v)$ solves problem (2).

After these preparations, the following theorem concludes this subsection by presenting results on solvability conditions for problem (1) and problem (2).

Theorem 1. Let $\mathcal{K}_\varepsilon$ be an ellipsoidal cone defined as in (3).

- (a) problem (1) has solutions for any given $A \in \mathbb{R}^{n \times n}$.
- (b) problem (2) has solutions if A is $\mathcal{K}_\varepsilon$ -regular and the triple (A, B, C) is $\mathcal{K}_\varepsilon$ -hyperbolic.
- (c) 0 is an eigenvalue of problem (2) if and only if $C \in R'_0(\mathcal{K}_\varepsilon)$.
- (d) There exists at least one positive and one negative eigenvalue of problem (2) if $C \in S'_0(\mathcal{K}_\varepsilon)$ and A is strictly $\mathcal{K}_\varepsilon$ -copositive.
- (e) There exists at least one positive and one negative eigenvalue of problem (2) if $A \in S'_0(\mathcal{K}_\varepsilon)$ and C is strictly $\mathcal{K}_\varepsilon$ -copositive.

Proof. (a) We first prove that the ellipsoidal cone $\mathcal{K}_\varepsilon$ defined as in (3) is pointed, i.e., $\mathcal{K}_\varepsilon \cap (-\mathcal{K}_\varepsilon) = \{0\}$. If $x \in \mathcal{K}_\varepsilon \cap (-\mathcal{K}_\varepsilon)$, then $x^T Qx \leq 0$ and $u_n^T x = 0$. From the eigenvalue decomposition of $Q := U\Lambda U^T$, where $U = [u_1, u_2, \dots, u_n]$ and $\Lambda = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$ with (λ_i, u_i) satisfying (5) and (6), there exists a vector $\alpha = [\alpha_1, \alpha_2, \dots, \alpha_n]^T \in \mathbb{R}^n$ such that $x = U\alpha$. Consequently, we obtain

$$u_n^T x = u_n^T \left(\sum_{i=1}^n \alpha_i u_i \right) = \alpha_n = 0, \quad x^T Qx = \alpha^T U^T Q U \alpha = \alpha^T \Lambda \alpha = \sum_{i=1}^n \lambda_i \alpha_i^2 = \sum_{i=1}^{n-1} \lambda_i \alpha_i^2 \leq 0.$$

Since $\lambda_i > 0$ ($i = 1, 2, \dots, n-1$), from the above inequality, we have $\alpha_i = 0$ ($i = 1, 2, \dots, n-1$). It follows from the relation $x = U\alpha$ and the fact $\alpha_i = 0$ ($i = 1, 2, \dots, n$) that x is equal to the null vector, which implies that $\mathcal{K}_\varepsilon$ is pointed. Next, the existence of a solution for problem (1) is proved. Recall that a nonempty set $S \subseteq \mathbb{R}^n$ is said to be a base for the cone $\mathcal{K}_\varepsilon$ if S does not contain the origin and

$\mathcal{K}_{\mathcal{E}} = \{\alpha p : \alpha \in \mathbb{R}_+, p \in S\}$. Suppose that $x \in \mathcal{K}_{\mathcal{E}} \setminus \{0\}$ and x solves problem (1), then necessarily $\lambda = \frac{\langle x, Ax \rangle}{\langle x, x \rangle}$. Consequently, problem (1) has a solution if and only if there exists a nonzero vector $x \in \mathcal{K}_{\mathcal{E}}$ such that

$$P_A(x) := Ax - \frac{\langle x, Ax \rangle}{\langle x, x \rangle} x \in \mathcal{K}_{\mathcal{E}}^*.$$

Since $\mathcal{K}_{\mathcal{E}}$ is a pointed closed convex cone and locally compact, it admits a compact convex base. Let S denote such a compact convex base for $\mathcal{K}_{\mathcal{E}}$, and define the mapping $\Phi : S \times S \rightarrow \mathbb{R}$ by

$$\Phi(x, p) = \langle P_A(x), p \rangle = \langle Ax, p \rangle - \frac{\langle x, Ax \rangle}{\langle x, x \rangle} \langle x, p \rangle.$$

- For any fixed $p \in S$, the function $\Phi(\cdot, p)$ is continuous, which follows from the fact that the functions $\langle Ax, p \rangle$, $\langle x, Ax \rangle$, $\langle x, x \rangle$, and $\langle x, p \rangle$ are continuous in x .
- For any fixed $x \in S$, the function $\Phi(x, \cdot)$ is linear. This follows since for any $p_1, p_2 \in S$ and $\lambda \in [0, 1]$:

$$\begin{aligned} & \Phi(x, \lambda p_1 + (1 - \lambda)p_2) \\ &= \langle Ax, \lambda p_1 + (1 - \lambda)p_2 \rangle - \frac{\langle x, Ax \rangle}{\langle x, x \rangle} \langle x, \lambda p_1 + (1 - \lambda)p_2 \rangle \\ &= \lambda \langle Ax, p_1 \rangle + (1 - \lambda) \langle Ax, p_2 \rangle - \lambda \frac{\langle x, Ax \rangle}{\langle x, x \rangle} \langle x, p_1 \rangle - (1 - \lambda) \frac{\langle x, Ax \rangle}{\langle x, x \rangle} \langle x, p_2 \rangle \\ &= \lambda \Phi(x, p_1) + (1 - \lambda) \Phi(x, p_2) \end{aligned}$$

- For any fixed $p \in S$, we also obtain

$$\Phi(p, p) = \langle Ap, p \rangle - \frac{\langle p, Ap \rangle}{\langle p, p \rangle} \langle p, p \rangle = 0.$$

From these three properties of the mapping $\Phi(x, p)$, Ky Fan's inequality in Lemma 3 guarantees the existence of $\bar{x} \in S \subseteq \mathcal{K}_{\mathcal{E}}$ such that $\Phi(\bar{x}, p) \geq 0$ for all $p \in S$, which yields that $P_A(\bar{x}) \in \mathcal{K}_{\mathcal{E}}^*$, since S constitutes a compact convex base for $\mathcal{K}_{\mathcal{E}}$.

(b) Building upon part (a), there exists a set $S \subseteq \mathbb{R}^n$ that is a compact convex base for $\mathcal{K}_{\mathcal{E}}$. Suppose that A is $\mathcal{K}_{\mathcal{E}}$ -regular and (A, B, C) is $\mathcal{K}_{\mathcal{E}}$ -hyperbolic. Then the quadratic equation

$$\lambda^2 \langle x, Ax \rangle + \lambda \langle x, Bx \rangle + \langle x, Cx \rangle = 0$$

admits real roots for any $x \in S$:

$$\lambda_{\pm}(x) = \frac{-\langle x, Bx \rangle \pm \sqrt{\langle x, Bx \rangle^2 - 4\langle x, Ax \rangle \langle x, Cx \rangle}}{2\langle x, Ax \rangle}.$$

Notice that these roots are well-defined on the set S . Consider the mapping $\Psi : S \times S \rightarrow \mathbb{R}$ defined by

$$\Psi(x, z) = \langle Q(\lambda_+(x))x, z \rangle, \quad Q(\lambda) := \lambda^2 A + \lambda B + C.$$

The following properties of the mapping $\Psi(x, z)$ hold.

- For any fixed $z \in S$, $\Psi(\cdot, z)$ is continuous. This follows because $\langle x, Ax \rangle$, $\langle x, Bx \rangle$, and $\langle x, Cx \rangle$ are continuous in x , implying $\lambda_+(x)$ and consequently $\Psi(\cdot, z)$ are continuous.
- For any fixed $x \in S$, $\Psi(x, \cdot)$ is linear. To verify this, fix $x \in S$ and observe that for any $z_1, z_2 \in \mathbb{R}^n$ and scalars $a, b \in \mathbb{R}$:

$$\begin{aligned} & \Psi(x, az_1 + bz_2) \\ &= \langle Q(\lambda_+(x))x, az_1 + bz_2 \rangle \\ &= a \langle Q(\lambda_+(x))x, z_1 \rangle + b \langle Q(\lambda_+(x))x, z_2 \rangle \\ &= a \Psi(x, z_1) + b \Psi(x, z_2). \end{aligned}$$

This confirms the linearity of $\Psi(x, \cdot)$.

- For any fixed $z \in S$, we also obtain

$$\Psi(z, z) = \langle z, Q(\lambda_+(z))z \rangle = \lambda_+(z)^2 \langle z, Az \rangle + \lambda_+(z) \langle z, Bz \rangle + \langle z, Cz \rangle = 0.$$

Similar to Part (a), Ky Fan's inequality in Lemma 3 guarantees the existence of $\bar{x} \in S \subseteq \mathcal{K}_{\mathcal{E}}$ such that $\Psi(\bar{x}, z) \geq 0$ for all $z \in S$, which implies that $(\lambda_+(\bar{x}), \bar{x})$ solves problem (2).

(c) If $C \in R'_0(\mathcal{K}_{\mathcal{E}})$, there exists at least one nonzero $x \in \mathcal{K}_{\mathcal{E}}$ satisfying $Cx \in \mathcal{K}_{\mathcal{E}}^*$ and $\langle x, Cx \rangle = 0$. It follows from the definition of problem (2) that 0 is an eigenvalue of problem (2). Conversely, if 0 is an eigenvalue of problem (2), then there exists a nonzero $x \in \mathbb{R}^n$ such that $x \in \mathcal{K}_{\mathcal{E}}$, $Cx \in \mathcal{K}_{\mathcal{E}}^*$, and $\langle x, Cx \rangle = 0$, which implies that $C \in R'_0(\mathcal{K}_{\mathcal{E}})$.

(d) Suppose that A is strictly $\mathcal{K}_{\mathcal{E}}$ -copositive and J is defined as in Lemma 4, which implies J is strictly $\mathcal{K}_{\mathcal{E}} \times \mathcal{K}_{\mathcal{E}}$ -copositive. Consider the model (7). If $y := (u, v) \in \mathcal{K}_{\mathcal{E}} \times \mathcal{K}_{\mathcal{E}}$ is a nonzero vector satisfying (7) and J is strictly $\mathcal{K}_{\mathcal{E}} \times \mathcal{K}_{\mathcal{E}}$ -copositive, then necessarily

$$\lambda = \frac{\langle y, Gy \rangle}{\langle y, Jy \rangle}.$$

Hence, the model (7) has a solution if and only if there exists a nonzero vector $y := (u, v) \in \mathcal{K}_\varepsilon \times \mathcal{K}_\varepsilon$ such that

$$P_{JG}(y) := \frac{\langle y, Gy \rangle}{\langle y, Jy \rangle} Jy - Gy \in \mathcal{K}_\varepsilon^* \times \mathcal{K}_\varepsilon^*.$$

Since $\mathcal{K}_\varepsilon$ is a pointed, locally compact, closed convex cone, $\mathcal{K}_\varepsilon \times \mathcal{K}_\varepsilon$ admits a compact convex base $\hat{S}$. Define the mapping $\hat{\Phi} : \hat{S} \times \hat{S} \rightarrow \mathbb{R}$ by

$$\hat{\Phi}(y, p) = \langle P_{JG}(y), p \rangle = \frac{\langle y, Gy \rangle}{\langle y, Jy \rangle} \langle Jy, p \rangle - \langle Gy, p \rangle.$$

Similar to Part (a), one can verify that (i) $\hat{\Phi}(\cdot, p)$ is continuous for any fixed $p \in \hat{S}$; (ii) $\hat{\Phi}(y, \cdot)$ is linear for any fixed $y \in \hat{S}$; (iii) $\hat{\Phi}(p, p) = 0$ for any fixed $p \in \hat{S}$. Consequently, Ky Fan's inequality in Lemma 3 guarantees the existence of $\hat{y} \in \hat{S} \subseteq \mathcal{K}_\varepsilon \times \mathcal{K}_\varepsilon$ such that $\hat{\Phi}(\hat{y}, p) \geq 0$ for all $p \in \hat{S}$, which implies that $\left(\frac{\langle \hat{y}, G\hat{y} \rangle}{\langle \hat{y}, J\hat{y} \rangle}, \hat{y} \right)$ solves the model (7). Thus, the model (7) has a solution if J is strictly $\mathcal{K}_\varepsilon \times \mathcal{K}_\varepsilon$ -copositive. Similarly, one can also deduce that the model (8) has a solution if J is strictly $\mathcal{K}_\varepsilon \times \mathcal{K}_\varepsilon$ -copositive. Next, we show that if $C \in S'_0(\mathcal{K}_\varepsilon)$, then 0 cannot be an eigenvalue of the model (7) or (8). Suppose 0 is an eigenvalue of the model (7) with corresponding eigenvector $0 \neq y = (u, v) \in \mathbb{R}^n \times \mathbb{R}^n$. This implies $Bu + Cv \in \mathcal{K}_\varepsilon^*$, $-Eu \in \mathcal{K}_\varepsilon^*$ and $u, v \in \mathcal{K}_\varepsilon$. It follows from the fact that $-\langle u, Eu \rangle \geq 0$ and the positive definiteness of E that $u = 0$. Consequently, $Cv \in \mathcal{K}_\varepsilon^*$ with $y \neq 0$ implying $v \neq 0$, contradicting $C \in S'_0(\mathcal{K}_\varepsilon)$. An identical argument applies to the model (8) as well. From Lemma 4 (b), the eigenvalues of the model (7) and the model (8) are nonzero and positive. Therefore, problem (2) admits at least one positive and one negative eigenvalue.

(e) Applying Part (d) to the following problem

$$x \in \mathcal{K}_\varepsilon, \lambda^2 Cx + \lambda Bx + Ax \in \mathcal{K}_\varepsilon^*, \langle x, \lambda^2 Cx + \lambda Bx + Ax \rangle = 0,$$

which yields a solution (λ, x) with $\lambda > 0$ satisfying

$$y = \lambda^2 Cx + \lambda Bx + Ax \in \mathcal{K}_\varepsilon^*, x \in \mathcal{K}_\varepsilon, \langle x, y \rangle = 0. \quad (9)$$

Dividing the first inclusion in (9) by λ^2 yields

$$\bar{y} = \mu^2 Ax + \mu Bx + Cx \in \mathcal{K}_\varepsilon^*, x \in \mathcal{K}_\varepsilon, \langle x, \bar{y} \rangle = 0,$$

establishing that (μ, x) solves problem (2) with $\mu := \lambda^{-1} > 0$. Applying the same reasoning to

$$x \in \mathcal{K}_\varepsilon, \lambda^2 Cx - \lambda Bx + Ax \in \mathcal{K}_\varepsilon^*, \langle x, \lambda^2 Cx - \lambda Bx + Ax \rangle = 0,$$

which produces a solution $(\bar{\lambda}, \bar{x})$ with $\bar{\lambda} > 0$ such that $(\bar{\lambda}^{-1}, \bar{x})$ solves

$$x \in \mathcal{K}_\varepsilon, \lambda^2 Ax - \lambda Bx + Cx \in \mathcal{K}_\varepsilon^*, \langle x, \lambda^2 Ax - \lambda Bx + Cx \rangle = 0.$$

Thus, $-\bar{\lambda}^{-1}$ constitutes a negative eigenvalue for problem (2). $\square$

Remark 1. While the solvability of eigenvalue problems over general cones has been discussed in the literature [34–36], our discussion of the ellipsoidal cone case serves to demonstrate how to construct the function φ in Lemma 3 to suit this specific geometry, an insight that may inform future research on other types of nonsymmetric cones.

3.2. Model reformulations

We develop model reformulations for problem (1) and problem (2) in this subsection. Since the null vector constitutes a trivial solution for both, we introduce the regularization constraint $\langle a, x \rangle = 1$ with $a \in \text{int}\mathcal{K}_\varepsilon^*$ to exclude this degenerate case. This yields the regularized problems

$$\begin{aligned} x \in \mathcal{K}_\varepsilon, Ax - \lambda x \in \mathcal{K}_\varepsilon^*, \langle x, Ax - \lambda x \rangle = 0, \langle a, x \rangle = 1; \\ x \in \mathcal{K}_\varepsilon, \lambda^2 Ax + \lambda Bx + Cx \in \mathcal{K}_\varepsilon^*, \langle x, \lambda^2 Ax + \lambda Bx + Cx \rangle = 0, \langle a, x \rangle = 1. \end{aligned}$$

By Lemma 1, there exist $\tilde{x}, \tilde{y} \in \mathcal{K}^n$ and $\tilde{a} \in \text{int}\mathcal{K}^n$ such that these problems reduce to

$$\tilde{x} \in \mathcal{K}^n, \tilde{y} = DU^T(A - \lambda I_n)UD\tilde{x} \in \mathcal{K}^n, \langle \tilde{x}, \tilde{y} \rangle = 0, \langle \tilde{a}, \tilde{x} \rangle = 1; \quad (10)$$

$$\tilde{x} \in \mathcal{K}^n, \tilde{y} = DU^T(\lambda^2 A + \lambda B + C)UD\tilde{x} \in \mathcal{K}^n, \langle \tilde{x}, \tilde{y} \rangle = 0, \langle \tilde{a}, \tilde{x} \rangle = 1, \quad (11)$$

where $U, D \in \mathbb{R}^{n \times n}$ are defined as in Lemma 1.

For any $\tilde{x} \in \mathbb{R}^n \setminus \{0\}$, problem (1) implies

$$\lambda = \frac{w^T \tilde{A}w}{w^T w}, \quad w := D\tilde{x}, \quad \tilde{A} := U^T A U \quad (12)$$

and consequently

$$\tilde{y} = D\tilde{A}D\tilde{x} - \lambda D^2\tilde{x} = D\tilde{A}D\tilde{x} - \frac{w^T \tilde{A}w}{w^T w} D^2\tilde{x}.$$

These expressions motivate the mapping $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$ defined by

$$F(\tilde{x}) := D\tilde{A}D\tilde{x} - \frac{w^T \tilde{A}w}{w^T w} D^2\tilde{x}, \quad (13)$$

which allows us to reformulate problem (1) as the following nonlinear second-order conic complementarity problem (NSOCCP)

$$\tilde{x} \in \mathcal{K}^n, \quad F(\tilde{x}) \in \mathcal{K}^n, \quad \langle \tilde{x}, F(\tilde{x}) \rangle = 0. \quad (14)$$

Notice that F is undefined at $\tilde{x} = 0$.

The following continuity result on the function F facilitates subsequent analysis.

Lemma 5. Consider problem (1), where $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is defined as in (13) and $\tilde{x} \neq 0_n$. Then

$$\lim_{\tilde{x} \rightarrow 0} F(\tilde{x}) = 0.$$

Proof. Given $\tilde{x} \neq 0$ and $w = D\tilde{x}$, the Rayleigh quotient satisfies

$$\begin{aligned} \left| \frac{w^T \tilde{A} w}{w^T w} \right| &\leq \frac{\sum_{i=1}^n \sum_{j=1}^n |\tilde{A}_{ij}| |w_i| |w_j|}{\sum_{i=1}^n w_i^2} \\ &\leq \left(\max_{1 \leq i, j \leq n} |\tilde{A}_{ij}| \right) \frac{\sum_{i=1}^n \sum_{j=1}^n \frac{w_i^2 + w_j^2}{2}}{\sum_{i=1}^n w_i^2} \\ &= \frac{\max_{i,j} |\tilde{A}_{ij}|}{2} \cdot \frac{n \sum_{i=1}^n w_i^2 + n \sum_{j=1}^n w_j^2}{\sum_{i=1}^n w_i^2} \\ &= n \max_{1 \leq i, j \leq n} |\tilde{A}_{ij}|. \end{aligned}$$

Combining this bound with the definition of F in (13) yields

$$\begin{aligned} \|F(\tilde{x})\| &\leq \|D\tilde{A}D\tilde{x}\| + \left| \frac{w^T \tilde{A} w}{w^T w} \right| \|D^2 \tilde{x}\| \\ &\leq \|D\tilde{A}D\tilde{x}\| + n \left(\max_{1 \leq i, j \leq n} |\tilde{A}_{ij}| \right) \|D^2 \tilde{x}\| \rightarrow 0, \text{ when } \tilde{x} \rightarrow 0, \end{aligned}$$

which completes the proof. $\square$

Remark 2. In general, studies on NSOCCP (14) require the differentiability of F . From Lemma 5, we can redefine $F(\tilde{x})$ as

$$F(\tilde{x}) := \begin{cases} D\tilde{A}D\tilde{x} - \frac{w^T \tilde{A} w}{w^T w} D^2 \tilde{x} & \text{if } \tilde{x} \neq 0, \\ 0 & \text{if } \tilde{x} = 0, \end{cases} \quad (15)$$

which makes $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$ continuous. Furthermore, $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is smooth except for 0. To see this, we define

$$\hat{F}(\tilde{x}) := \begin{cases} \frac{w^T \tilde{A} w}{w^T w} D^2 \tilde{x} & \text{if } \tilde{x} \neq 0, \\ 0 & \text{if } \tilde{x} = 0. \end{cases}$$

We only need to show that $\hat{F}$ is not differentiable at 0. We construct the following example, in which $n = 2$ and the parameters Q, u_n in (3) are set to be

$$Q = \begin{bmatrix} 1 & 0 \\ 0 & -\tan^2 \theta \end{bmatrix}, \quad u_n = e_n.$$

The corresponding ellipsoidal cone reduces to a circular cone with the matrices U and D given by

$$U = I_2, \quad D = \begin{bmatrix} 1 & 0 \\ 0 & (\tan^2 \theta)^{-\frac{1}{2}} \end{bmatrix}.$$

For $n = 2$, we take $A = \begin{bmatrix} A_{11} & 0 \\ 0 & A_{22} \end{bmatrix}$, where $A_{11} \neq A_{22}$. For arbitrary $\tilde{x} = (\tilde{x}_1, \tilde{x}_2) \in \mathbb{R}^2$, we can rewrite $\hat{F}$ as

$$\hat{F}(\tilde{x}) := \begin{bmatrix} \hat{F}_1(\tilde{x}) \\ \hat{F}_2(\tilde{x}) \end{bmatrix} = \frac{A_{11} \tilde{x}_1^2 + A_{22} (\tan^2 \theta)^{-1} \tilde{x}_2^2}{\tilde{x}_1^2 + (\tan^2 \theta)^{-1} \tilde{x}_2^2} \begin{bmatrix} \tilde{x}_1 \\ (\tan^2 \theta)^{-1} \tilde{x}_2 \end{bmatrix},$$

and

$$\begin{aligned} \frac{\partial \hat{F}_1}{\partial \tilde{x}_1}(0) &= \lim_{\tilde{x}_1 \rightarrow 0} \frac{\hat{F}_1(\tilde{x}_1, 0) - \hat{F}_1(0, 0)}{\tilde{x}_1} = \lim_{\tilde{x}_1 \rightarrow 0} \frac{A_{11} \tilde{x}_1}{\tilde{x}_1} = A_{11}, \\ \frac{\partial \hat{F}_1}{\partial \tilde{x}_2}(0) &= \lim_{\tilde{x}_2 \rightarrow 0} \frac{\hat{F}_1(0, \tilde{x}_2) - \hat{F}_1(0, 0)}{\tilde{x}_2} = \lim_{\tilde{x}_2 \rightarrow 0} \frac{A_{22} \cdot 0}{\tilde{x}_2} = 0. \end{aligned}$$

Now, we consider

$$\lim_{\tilde{x} \rightarrow 0} \frac{\hat{F}_1(\tilde{x}) - \hat{F}_1(0) - \left(\frac{\partial \hat{F}_1}{\partial \tilde{x}_1}(0), \frac{\partial \hat{F}_1}{\partial \tilde{x}_2}(0) \right) \tilde{x}}{\|\tilde{x}\|},$$

and simplify it as

$$\lim_{\tilde{x} \rightarrow 0} \frac{(A_{22} - A_{11})(\tan^2 \theta)^{-1} \tilde{x}_1 \tilde{x}_2^2}{(\tilde{x}_1^2 + (\tan^2 \theta)^{-1} \tilde{x}_2^2)(\tilde{x}_1^2 + \tilde{x}_2^2)^{\frac{1}{2}}}.$$

If we choose $\tilde{x}_1 = \tilde{x}_2$, then the above equation becomes

$$\frac{(A_{22} - A_{11})(\tan^2 \theta)^{-1}}{\sqrt{2}(1 + (\tan^2 \theta)^{-1})} \neq 0.$$

This implies that $\hat{F}_1$ is not differentiable at 0 and hence F is not differentiable at 0.

From Remark 2 and Lemma 2, the next lemma presents the explicit formulation of ∇F at $\tilde{x} \neq 0$.

Lemma 6. Consider the model (10), where $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is defined as in (15); the mapping F is smooth except for 0 and its gradient is expressed as

$$\nabla F(\tilde{x}) = D\tilde{A}^T D - [D(\tilde{A} + \tilde{A}^T)w w^T - 2(w^T \tilde{A} w)D] \frac{w w^T D}{(w^T w)^2} - \frac{w^T \tilde{A} w}{w^T w} D^2.$$

Proof. Denote by $f(\tilde{x}) := \frac{w^T \tilde{A} w}{w^T w}$ and $G(\tilde{x}) = D^2 \tilde{x}$. Then we have

$$F(\tilde{x}) = D\tilde{A}D\tilde{x} - f(\tilde{x})G(\tilde{x}).$$

It follows from Lemma 2 and $\tilde{x} \neq 0$ that

$$\nabla F(\tilde{x}) = D\tilde{A}^T D - \nabla f(\tilde{x})(G(\tilde{x}))^T - f(\tilde{x})\nabla G(\tilde{x}),$$

where $\nabla f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ has the following expression

$$\begin{aligned} \nabla f(\tilde{x}) &= \frac{\nabla(w^T \tilde{A} w) \cdot (w^T w) - (w^T \tilde{A} w) \cdot \nabla(w^T w)}{(w^T w)^2} \\ &= \frac{D(\tilde{A} + \tilde{A}^T)D\tilde{x} \cdot (w^T w) - (w^T \tilde{A} w) \cdot 2D^2 \tilde{x}}{(w^T w)^2} \\ &= \frac{[D(\tilde{A} + \tilde{A}^T)w w^T - 2(w^T \tilde{A} w)D]w}{(w^T w)^2}. \end{aligned}$$

In addition, we also obtain $(G(\tilde{x}))^T = (D^2 \tilde{x})^T = (Dw)^T = w^T D$ and $\nabla G(\tilde{x}) = D^2$. $\square$

Now, we summarize the relation between the ellipsoidal conic eigenvalue complementarity problem (10) and the nonlinear second-order conic complementarity problem (14) in the following theorem.

Theorem 2. Consider the model (10), where $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is defined as in (15). Then the following relations hold.

- (a) If $(\tilde{x}^*, \lambda^*)$ solves (10), then $\tilde{x}^*$ solves (14).
 (b) Conversely, if $\tilde{x}$ is a nonzero solution of (14), then $(\tilde{x}^*, \lambda^*)$ solves (10), where

$$\lambda^* = \frac{\tilde{w}^T \tilde{A} \tilde{w}}{\tilde{w}^T \tilde{w}}, \quad \tilde{x}^* = \frac{\tilde{x}}{\tilde{a}^T \tilde{x}}, \quad \tilde{w} = D\tilde{x}.$$

Moreover, because $\tilde{x} \in \mathcal{K}^n \setminus \{0\}$ and $\tilde{a} \in \text{int } \mathcal{K}^n$, we have $\tilde{a}^T \tilde{x} > 0$, so the scaling is well-defined and does not produce the zero solution.

Proof. It follows from the above discussion that the conclusion of Part (a) is trivial. Hence, we only need to prove Part (b). Suppose that $\tilde{x}$ is a nonzero solution of (14) and $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is defined as in (15). Then we have $\tilde{w} = D\tilde{x} \neq 0$ and

$$\tilde{x} \in \mathcal{K}^n, \quad D\tilde{A}D\tilde{x} - \frac{\tilde{w}^T \tilde{A} \tilde{w}}{\tilde{w}^T \tilde{w}} D^2 \tilde{x} \in \mathcal{K}^n, \quad \left\langle \tilde{x}, D\tilde{A}D\tilde{x} - \frac{\tilde{w}^T \tilde{A} \tilde{w}}{\tilde{w}^T \tilde{w}} D^2 \tilde{x} \right\rangle = 0.$$

Since $\tilde{a} := (\tilde{a}_1, \tilde{a}_2) \in \text{int } \mathcal{K}^n$ and $0 \neq \tilde{x} = (\tilde{x}_1, \tilde{x}_2) \in \mathcal{K}^n$, we obtain $\tilde{x}_2 > 0$ and

$$\langle \tilde{a}, \tilde{x} \rangle = \langle \tilde{a}_1, \tilde{x}_1 \rangle + \tilde{a}_2 \tilde{x}_2 > |\langle \tilde{a}_1, \tilde{x}_1 \rangle| + \langle \tilde{a}_1, \tilde{x}_1 \rangle \geq 0,$$

which implies that the scaling is well-defined. From all the above, we conclude that

$$\begin{aligned} \tilde{x}^* &= \frac{1}{\tilde{a}^T \tilde{x}} \tilde{x} \in \mathcal{K}^n, \quad \tilde{y}^* = D\tilde{A}D\tilde{x}^* - \lambda^* D^2 \tilde{x}^* = \frac{1}{\tilde{a}^T \tilde{x}} \left[D\tilde{A}D\tilde{x} - \frac{\tilde{w}^T \tilde{A} \tilde{w}}{\tilde{w}^T \tilde{w}} D^2 \tilde{x} \right] \in \mathcal{K}^n, \\ (\tilde{x}^*)^T \tilde{y}^* &= \left(\frac{1}{\tilde{a}^T \tilde{x}} \right)^2 \left\langle \tilde{x}, D\tilde{A}D\tilde{x} - \frac{\tilde{w}^T \tilde{A} \tilde{w}}{\tilde{w}^T \tilde{w}} D^2 \tilde{x} \right\rangle = 0, \quad \tilde{a}^T \tilde{x}^* = \frac{\tilde{a}^T \tilde{x}}{\tilde{a}^T \tilde{x}} = 1, \end{aligned}$$

which show that $(\tilde{x}^*, \lambda^*)$ solves (10) and $\tilde{x}^* \neq 0$. $\square$

Next, we turn to the quadratic case (11). Assume that the matrix $A \in \mathbb{R}^{n \times n}$ is positive definite (hence A is $\mathcal{K}_\varepsilon$ -regular) and the triple (A, B, C) is $\mathcal{K}_\varepsilon$ -hyperbolic. For any given $\tilde{x} \in \mathbb{R}^n$ and $\tilde{x} \neq 0$, it follows from (11) that

$$(w^T \tilde{A}w)\lambda^2 + (w^T \tilde{B}w)\lambda + w^T \tilde{C}w = 0, \quad (16)$$

where $\tilde{A} := U^T A U$, $\tilde{B} := U^T B U$, $\tilde{C} := U^T C U$ and $w := D\tilde{x} \neq 0$. From the Eq. (16), we obtain

$$\lambda_1(\tilde{x}) = \frac{-w^T \tilde{B}w + \sqrt{(w^T \tilde{B}w)^2 - 4(w^T \tilde{A}w)(w^T \tilde{C}w)}}{2(w^T \tilde{A}w)}, \quad (17)$$

$$\lambda_2(\tilde{x}) = \frac{-w^T \tilde{B}w - \sqrt{(w^T \tilde{B}w)^2 - 4(w^T \tilde{A}w)(w^T \tilde{C}w)}}{2(w^T \tilde{A}w)}, \quad (18)$$

which also motivates the mapping $F_i : \mathbb{R}^n \rightarrow \mathbb{R}^n$ defined by

$$F_i(\tilde{x}) := \lambda_i^2(\tilde{x}) D \tilde{A} D \tilde{x} + \lambda_i(\tilde{x}) D \tilde{B} D \tilde{x} + D \tilde{C} D \tilde{x} \quad (i = 1, 2), \quad (19)$$

where $\lambda_1(\tilde{x})$ and $\lambda_2(\tilde{x})$ are defined as in (17) and (18).

Similar to Lemma 5, the continuity result on the functions F_i are established in the following lemma.

Lemma 7. Consider the model (11), where $F_i : \mathbb{R}^n \rightarrow \mathbb{R}^n$ ($i = 1, 2$) is defined as in (19) and $\tilde{x} \neq 0$. Then

$$\lim_{\tilde{x} \rightarrow 0} F_i(\tilde{x}) = 0.$$

Proof. From the fact that $A \in \mathbb{R}^{n \times n}$ is positive definite and by the Cholesky factorization of A , there exists an invertible lower triangular matrix L with positive diagonal entries such that $A = LL^T$ and the following relations are satisfied, i.e.,

$$\begin{aligned} w^T \tilde{A}w &= (L^T U w)^T (L^T U w), \\ w^T \tilde{B}w &= (L^T U w)^T (L^{-1} B (L^{-1})^T) (L^T U w), \\ w^T \tilde{C}w &= (L^T U w)^T (L^{-1} C (L^{-1})^T) (L^T U w). \end{aligned}$$

For convenience, we denote

$$E := L^{-1} B (L^{-1})^T, \quad F := L^{-1} C (L^{-1})^T, \quad z := L^T U w,$$

$$M_1 := \max_{1 \leq i, j \leq n} |e_{ij}|, \quad M_2 := \max_{1 \leq i, j \leq n} |f_{ij}|,$$

where e_{ij} is the (i, j) -element of E , f_{ij} is the (i, j) -element of F . Similar to the proof of Lemma 5, we also obtain

$$\begin{aligned} |w^T \tilde{A}w| &= |z^T z| = \sum_{i=1}^n z_i^2, \\ |w^T \tilde{B}w| &= |z^T E z| \leq n M_1 \sum_{i=1}^n z_i^2, \\ |w^T \tilde{C}w| &= |z^T F z| \leq n M_2 \sum_{i=1}^n z_i^2. \end{aligned}$$

Hence, for each i and $\tilde{x} \neq 0$, we have

$$\begin{aligned} |\lambda_i(\tilde{x})| &\leq \frac{|w^T \tilde{B}w| + \sqrt{|w^T \tilde{B}w|^2 + 4|w^T \tilde{A}w||w^T \tilde{C}w|}}{2|w^T \tilde{A}w|} \\ &\leq \frac{n M_1 \sum_{i=1}^n z_i^2 + \sqrt{(n M_1 \sum_{i=1}^n z_i^2)^2 + 4(\sum_{i=1}^n z_i^2)(n M_2 \sum_{i=1}^n z_i^2)}}{2 \sum_{i=1}^n z_i^2} \\ &\leq \frac{1}{2} n \max\{M_1, M_2\} + \frac{1}{2} \sqrt{n^2 \max\{M_1, M_2\}^2 + 4n \max\{M_1, M_2\}} \\ &\leq n \max\{M_1, M_2\} + 1 := M_3. \end{aligned}$$

It follows from (19) and the above inequality that

$$\|F_i(\tilde{x})\| \leq M_3^2 \|D \tilde{A} D \tilde{x}\| + M_3 \|D \tilde{B} D \tilde{x}\| + \|D \tilde{C} D \tilde{x}\|,$$

which completes the proof. $\square$

Remark 3. From Lemma 7, we also redefine $F_i(\tilde{x})$ as

$$F_i(\tilde{x}) = \begin{cases} \lambda_i^2(\tilde{x}) D \tilde{A} D \tilde{x} + \lambda_i(\tilde{x}) D \tilde{B} D \tilde{x} + D \tilde{C} D \tilde{x} & \text{if } \tilde{x} \neq 0, \\ 0 & \text{if } \tilde{x} = 0, \end{cases} \quad (20)$$

where $\lambda_i(\tilde{x})$ ($i = 1, 2$) are defined as in (17) and (18), which makes $F_i : \mathbb{R}^n \rightarrow \mathbb{R}^n$ continuous. In addition, similar to Remark 2, we construct the following example, in which $n = 2$ and the parameters Q , u_n in (3) are set to be

$$Q = \begin{bmatrix} 1 & 0 \\ 0 & -\tan^2 \theta \end{bmatrix}, \quad u_n = e_n.$$

In this case, the corresponding ellipsoidal cone reduces to a circular cone with the matrices U and D given by

$$U = I_2, \quad D = \begin{bmatrix} 1 & 0 \\ 0 & (\tan^2 \theta)^{-\frac{1}{2}} \end{bmatrix}.$$

For $n = 2$, we take

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} 6 & 0 \\ 0 & 4 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix},$$

where A is positive definite and the triple (A, B, C) is $\mathcal{K}_\varepsilon$ -hyperbolic. Note that $F_i(\bar{x})$ as defined in (19) can be rewritten as

$$F_i(w) := \lambda_i^2(w) D \tilde{A} w + \lambda_i(w) D \tilde{B} w + D \tilde{C} w \quad (i = 1, 2).$$

Now, we consider the first term of $F_i(w)$

$$\begin{bmatrix} g_{i1}(w) \\ g_{i2}(w) \end{bmatrix} := \lambda_i^2(w) \begin{bmatrix} w_1 \\ 2(\tan^2 \theta)^{-\frac{1}{2}} w_2 \end{bmatrix}.$$

Denote by $g_{i1} : \mathbb{R}^2 \rightarrow \mathbb{R}$ the first line of the above expression. For arbitrary $w = (w_1, w_2) \in \mathbb{R}^2$, we have

$$\begin{aligned} \frac{\partial g_{i1}}{\partial w_1}(0) &= \lim_{w_1 \rightarrow 0} \frac{g_{i1}(w_1, 0) - g_{i1}(0, 0)}{w_1} = \lim_{w_1 \rightarrow 0} \frac{\lambda_i^2(w_1, 0) - \lambda_i^2(0, 0)}{w_1} = \lambda_i^2(w_1, 0), \\ \frac{\partial g_{i1}}{\partial w_2}(0) &= \lim_{w_2 \rightarrow 0} \frac{g_{i1}(0, w_2) - g_{i1}(0, 0)}{w_2} = \lim_{w_2 \rightarrow 0} \frac{\lambda_i^2(0, w_2) \cdot 0}{w_2} = 0. \end{aligned}$$

Now, we consider

$$\lim_{w \rightarrow 0} \frac{g_{i1}(w) - g_{i1}(0) - \left(\frac{\partial g_{i1}}{\partial w_1}(0) w_1 + \frac{\partial g_{i1}}{\partial w_2}(0) w_2 \right)}{\|w\|} = \lim_{w \rightarrow 0} \frac{\lambda_i^2(w_1, w_2) w_1 - \lambda_i^2(w_1, 0) w_1}{(w_1^2 + w_2^2)^{\frac{1}{2}}},$$

if we choose $w_1 = w_2 = t$ ($t > 0$), i.e. $\tilde{x}_1 = \tilde{x}_2(\tan^2 \theta)^{-\frac{1}{2}} = t$, we can simplify it as

$$\frac{1}{\sqrt{2}} (\lambda_i^2(t, t) - \lambda_i^2(t, 0)),$$

and we can calculate that $\lambda_1(t, t) = -\frac{1}{3}$, $\lambda_2(t, t) = -3$, $\lambda_1(t, 0) = -3 + 2\sqrt{2}$ and $\lambda_2(t, 0) = -3 - 2\sqrt{2}$, so the above expression is not equal to 0. This implies that g_{i1} is not differentiable at 0 and hence F_i is not differentiable at 0. Furthermore, because the functions $\lambda_1(\bar{x})$ and $\lambda_2(\bar{x})$ defined as in (17) and (18) contain the square root

$$\sqrt{(w^T \tilde{B} w)^2 - 4(w^T \tilde{A} w)(w^T \tilde{C} w)}.$$

Consequently, the differentiability of F_i may fail at points where the above discriminant is zero. Hence, we need a stronger assumption for the quadratic case (11). For example, we assume

$$\sqrt{(w^T \tilde{B} w)^2 - 4(w^T \tilde{A} w)(w^T \tilde{C} w)} > 0, \quad \text{for all } w \neq 0 \text{ and } w = D\tilde{x}. \quad (21)$$

From Remark 3 and Lemma 7, the next lemma presents the explicit formulation of ∇F_i at $\tilde{x} \neq 0$.

Lemma 8. Consider the model (11), where the matrix $A \in \mathbb{R}^{n \times n}$ is positive definite and the triple (A, B, C) is $\mathcal{K}_\varepsilon$ -hyperbolic. Suppose that $F_i : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is defined as in (20). Then the mapping F_i is smooth at every $\tilde{x} \neq 0$ such that the vector $w = D\tilde{x}$ satisfies assumption (21). Their gradients are expressed as

$$\nabla F_i(\tilde{x}) = \nabla \lambda_i(\tilde{x}) (2\lambda_i(\tilde{x}) \tilde{x}^T D \tilde{A}^T D + \tilde{x}^T D \tilde{B}^T D) + \lambda_i^2(\tilde{x}) D \tilde{A}^T D + \lambda_i(\tilde{x}) D \tilde{B}^T D + D \tilde{C}^T D.$$

where

$$\begin{aligned} \nabla \lambda_1(\tilde{x}) &= \frac{1}{2(w^T \tilde{A} w)} [D(\tilde{B} + \tilde{B}^T)w] \left[(G(\tilde{x}))^{-\frac{1}{2}} (w^T \tilde{B} w) - 1 \right] \\ &\quad - \frac{1}{(w^T \tilde{A} w)} (G(\tilde{x}))^{-\frac{1}{2}} \left[[D(\tilde{A} + \tilde{A}^T)w] \cdot (w^T \tilde{C} w) + [D(\tilde{C} + \tilde{C}^T)w] \cdot (w^T \tilde{A} w) \right] \\ &\quad + \frac{1}{2(w^T \tilde{A} w)^2} \left[(w^T \tilde{B} w) - \sqrt{G(\tilde{x})} \right] [D(\tilde{A} + \tilde{A}^T)w], \\ \nabla \lambda_2(\tilde{x}) &= -\frac{1}{2(w^T \tilde{A} w)} [D(\tilde{B} + \tilde{B}^T)w] \left[(G(\tilde{x}))^{-\frac{1}{2}} (w^T \tilde{B} w) + 1 \right] \\ &\quad + \frac{1}{(w^T \tilde{A} w)} (G(\tilde{x}))^{-\frac{1}{2}} \left[[D(\tilde{A} + \tilde{A}^T)w] \cdot (w^T \tilde{C} w) + [D(\tilde{C} + \tilde{C}^T)w] \cdot (w^T \tilde{A} w) \right] \\ &\quad + \frac{1}{2(w^T \tilde{A} w)^2} \left[(w^T \tilde{B} w) + \sqrt{G(\tilde{x})} \right] [D(\tilde{A} + \tilde{A}^T)w], \\ G(\tilde{x}) &:= (w^T \tilde{B} w)^2 - 4(w^T \tilde{A} w)(w^T \tilde{C} w). \end{aligned}$$

Proof. For convenience, we define the mapping $G : \mathbb{R}^n \rightarrow \mathbb{R}$ as

$$G(\tilde{x}) := (w^T \tilde{B}w)^2 - 4(w^T \tilde{A}w)(w^T \tilde{C}w).$$

From the definitions of $\lambda_i(\tilde{x})$ and the chain rule of gradient, we obtain

$$\begin{aligned} \nabla \lambda_1(\tilde{x}) &= \frac{1}{2(w^T \tilde{A}w)^2} \left[-\nabla(w^T \tilde{B}w) + \nabla \sqrt{G(\tilde{x})} \right] (w^T \tilde{A}w) \\ &\quad - \frac{1}{2(w^T \tilde{A}w)^2} \left[-(w^T \tilde{B}w) + \sqrt{G(\tilde{x})} \right] \nabla(w^T \tilde{A}w) \\ &= \frac{1}{2(w^T \tilde{A}w)} [D(\tilde{B} + \tilde{B}^T)w] \left[(G(\tilde{x}))^{-\frac{1}{2}} (w^T \tilde{B}w) - 1 \right] \\ &\quad - \frac{1}{(w^T \tilde{A}w)} (G(\tilde{x}))^{-\frac{1}{2}} \left[[D(\tilde{A} + \tilde{A}^T)w] \cdot (w^T \tilde{C}w) + [D(\tilde{C} + \tilde{C}^T)w] \cdot (w^T \tilde{A}w) \right] \\ &\quad + \frac{1}{2(w^T \tilde{A}w)^2} \left[(w^T \tilde{B}w) - \sqrt{G(\tilde{x})} \right] [D(\tilde{A} + \tilde{A}^T)w] \end{aligned}$$

and

$$\begin{aligned} \nabla \lambda_2(\tilde{x}) &= -\frac{1}{2(w^T \tilde{A}w)} [D(\tilde{B} + \tilde{B}^T)w] \left[(G(\tilde{x}))^{-\frac{1}{2}} (w^T \tilde{B}w) + 1 \right] \\ &\quad + \frac{1}{(w^T \tilde{A}w)} (G(\tilde{x}))^{-\frac{1}{2}} \left[[D(\tilde{A} + \tilde{A}^T)w] \cdot (w^T \tilde{C}w) + [D(\tilde{C} + \tilde{C}^T)w] \cdot (w^T \tilde{A}w) \right] \\ &\quad + \frac{1}{2(w^T \tilde{A}w)^2} \left[(w^T \tilde{B}w) + \sqrt{G(\tilde{x})} \right] [D(\tilde{A} + \tilde{A}^T)w]. \end{aligned}$$

In light of Lemma 2 for each i , we have

$$\begin{aligned} \nabla F_i(\tilde{x}) &= 2\lambda_i(\tilde{x}) \nabla \lambda_i(\tilde{x}) \cdot (\tilde{x}^T D \tilde{A}^T D) + \lambda_i^2(\tilde{x}) D \tilde{A} D + \nabla \lambda_i(\tilde{x}) \cdot (\tilde{x}^T D \tilde{B}^T D) \\ &\quad + \lambda_i(\tilde{x}) D \tilde{B}^T D + D \tilde{C}^T D \\ &= \nabla \lambda_i(\tilde{x}) (2\lambda_i(\tilde{x}) \tilde{x}^T D \tilde{A}^T D + \tilde{x}^T D \tilde{B}^T D) + \lambda_i^2(\tilde{x}) D \tilde{A}^T D + \lambda_i(\tilde{x}) D \tilde{B}^T D + D \tilde{C}^T D, \end{aligned}$$

which completes the proof. $\square$

The following theorem concludes this section by analyzing the relation between the ellipsoidal conic eigenvalue complementarity problem (11) and the nonlinear second-order conic complementarity problem (14).

Theorem 3. Consider the model (11), where $F_i : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is defined as in (20) and the matrix $A \in \mathbb{R}^{n \times n}$ is positive definite. Then the following relations hold.

- (a) If $(\tilde{x}^*, \lambda^*)$ solves (11), then $\tilde{x}^*$ solves (14) with $F = F_1$ or $F = F_2$.
 (b) Conversely, if $\tilde{x}$ is a nonzero solution of (14) with $F = F_1$ or $F = F_2$, then $(\tilde{x}^*, \lambda^*)$ solves (11), where $\tilde{x}^* = \frac{\tilde{x}}{\tilde{a}^T \tilde{x}}$ and $\lambda^* = \lambda_i(\tilde{x})$ with $\lambda_i(\tilde{x})$ ($i = 1, 2$) defined as in (17) and (18). Moreover, because $\tilde{x} \in \mathcal{K}^n \setminus \{0\}$ and $\tilde{a} \in \text{int } \mathcal{K}^n$, we have $\tilde{a}^T \tilde{x} > 0$, so the scaling is well-defined and does not produce the zero solution.

Proof. It follows from the above discussion that the conclusion of Part (a) is trivial. Hence, we only need to prove Part (b). Suppose that $\tilde{x}$ is a nonzero solution of (14) with $F = F_1$ or $F = F_2$. Then, we have

$$\begin{aligned} \tilde{x} \in \mathcal{K}^n, \quad \lambda_i^2(\tilde{x}) D \tilde{A} D \tilde{x} + \lambda_i(\tilde{x}) D \tilde{B} D \tilde{x} + D \tilde{C} D \tilde{x} \in \mathcal{K}^n, \\ \langle \tilde{x}, \lambda_i^2(\tilde{x}) D \tilde{A} D \tilde{x} + \lambda_i(\tilde{x}) D \tilde{B} D \tilde{x} + D \tilde{C} D \tilde{x} \rangle = 0. \end{aligned}$$

Similar to Theorem 2, we also obtain $\langle \tilde{a}, \tilde{x} \rangle > 0$ and consequently the scaling is well-defined. From all the above, we conclude that

$$\begin{aligned} \tilde{x}^* = \frac{1}{\tilde{a}^T \tilde{x}} \tilde{x} \in \mathcal{K}^n, \quad \tilde{y}^* = \frac{1}{\tilde{a}^T \tilde{x}} (\lambda_i^2(\tilde{x}) D \tilde{A} D \tilde{x} + \lambda_i(\tilde{x}) D \tilde{B} D \tilde{x} + D \tilde{C} D \tilde{x}) \in \mathcal{K}^n, \\ (\tilde{x}^*)^T \tilde{y}^* = \left(\frac{1}{\tilde{a}^T \tilde{x}} \right)^2 \langle \tilde{x}, \lambda_i^2(\tilde{x}) D \tilde{A} D \tilde{x} + \lambda_i(\tilde{x}) D \tilde{B} D \tilde{x} + D \tilde{C} D \tilde{x} \rangle = 0, \quad \tilde{a}^T \tilde{x}^* = \frac{\tilde{a}^T \tilde{x}}{\tilde{a}^T \tilde{x}} = 1, \end{aligned}$$

which show that $(\tilde{x}^*, \lambda^*)$ solves (11) and $\tilde{x}^* \neq 0$. $\square$

Remark 4. The reformulation pipelines for problem (1) and problem (2) established in Theorems 2 and 3 rely on the spectral decomposition $Q = U \Lambda U^T$ and the transformation matrix $T = U D$, where U and D are defined as in Lemma 1. For large n , this decomposition has $O(n^3)$ cost, which is a one-time preprocessing overhead. In many applications, Q may have special structure (e.g., sparsity). However, the transformation T generally destroys sparsity, making the transformed matrices $\tilde{A} = U^T A U$, $\tilde{B} = U^T B U$, $\tilde{C} = U^T C U$ fully dense. This is a potential limitation of the approach. For large-scale problems, one might avoid the explicit transformation and instead work directly with the ellipsoidal cone using tailored algorithms, however, the projection onto an ellipsoidal cone does not admit a closed-form expression, in contrast to the second-order cone case; see [1, Theorem 4.1] for more details.

4. Algorithm and numerical experiments

In this section we develop a projection (fixed-point) method for the NSOCCP (14) and report its numerical performance on synthetic instances of problems (1) and (2).

4.1. A projection method for problem (14)

By Theorems 2 and 3, problems (1) and (2) can be reformulated as NSOCCPs. For problem (1) we solve the NSOCCP (14) with F as in (15) by the classical projection (fixed-point) iteration

$$\tilde{x}^{k+1} = \Pi_{\mathcal{K}^n}(\tilde{x}^k - \alpha_k F(\tilde{x}^k)), \quad (22)$$

for problem (2) we solve, instead of the nonlinear NSOCCP (14) with F_i as in (20), the equivalent lifted linear model described at the end of this subsection. The projection onto the second-order cone $\mathcal{K}^n$ admits the explicit formula

$$\Pi_{\mathcal{K}^n}(x) = \begin{cases} x, & \|\tilde{x}\| \leq x_n, \\ 0, & \|\tilde{x}\| \leq -x_n, \\ \frac{1}{2} \left(1 + \frac{x_n}{\|\tilde{x}\|}\right)(\tilde{x}, \|\tilde{x}\|)^T, & \text{otherwise,} \end{cases}$$

where $x := (\tilde{x}, x_n) \in \mathbb{R}^{n-1} \times \mathbb{R}$. One can refer to [3, Proposition 3.3] for more details.

To exclude the trivial solution $\tilde{x} = 0$, which always satisfies (14) because $F(0) = 0$ and $F_i(0) = 0$ (cf. Proposition 2), we enforce the normalization $\langle \tilde{a}, \tilde{x} \rangle = 1$ of the regularized models (10) and (11) with $\tilde{a} = e_n$, i.e., $\tilde{x}_n = 1$. The feasible set then becomes the compact slice

$$S := \{\tilde{x} \in \mathcal{K}^n : \tilde{x}_n = 1\} = \{(\tilde{x}, 1) : \|\tilde{x}\| \leq 1\},$$

whose projection is given by

$$\Pi_S(z) = \left(\frac{\tilde{z}}{\max\{1, \|\tilde{z}\|\}}, 1 \right), \quad z = (\tilde{z}, z_n).$$

In the experiments, Algorithm 1 is run from several random points on S (multi-start), and the point with the smallest final residual is reported.

In Algorithm 1, the projection onto the cone in (22) is replaced by the projection onto the slice Π_S , which enforces the normalization $\tilde{x}_n = 1$; the stopping test remains the cone natural-residual test $\|R_{\alpha_0}(\tilde{x}^k)\| \leq \text{tol}_1$ of Proposition 1, so its exactness is unaffected.

The following proposition provides the exact reformulation on which the iteration (22) is based.

Proposition 1. Let $\alpha > 0$ be arbitrary. Define the natural residual $R_\alpha : \mathbb{R}^n \rightarrow \mathbb{R}^n$ by

$$R_\alpha(\tilde{x}) := \tilde{x} - \Pi_{\mathcal{K}^n}(\tilde{x} - \alpha F(\tilde{x})), \quad (23)$$

and the merit function $\Theta_\alpha(\tilde{x}) := \frac{1}{2} \|R_\alpha(\tilde{x})\|^2$. Then $\tilde{x}$ solves the NSOCCP (14) if and only if $R_\alpha(\tilde{x}) = 0$, i.e., if and only if $\Theta_\alpha(\tilde{x}) = 0$. In particular, Θ_α is a complementarity merit function for (14).

Proof. We use the projection characterization for the second-order cone $\mathcal{K}^n$:

$$z = \Pi_{\mathcal{K}^n}(w) \iff z \in \mathcal{K}^n, \quad w - z \in (\mathcal{K}^n)^\circ, \quad \langle w - z, z \rangle = 0,$$

where $(\mathcal{K}^n)^\circ := \{v : \langle v, y \rangle \leq 0 \text{ for all } y \in \mathcal{K}^n\}$ is the polar cone. Since $\mathcal{K}^n$ is self-dual, we have $(\mathcal{K}^n)^\circ = -\mathcal{K}^n$. If $\tilde{x}$ solves (14), then $\tilde{x} \in \mathcal{K}^n$, $F(\tilde{x}) \in \mathcal{K}^n$ and $\langle \tilde{x}, F(\tilde{x}) \rangle = 0$. Applying the characterization with $z = \tilde{x}$ and $w = \tilde{x} - \alpha F(\tilde{x})$, we obtain $w - z = -\alpha F(\tilde{x}) \in -\mathcal{K}^n = (\mathcal{K}^n)^\circ$ and $\langle w - z, z \rangle = -\alpha \langle F(\tilde{x}), \tilde{x} \rangle = 0$, whence $\Pi_{\mathcal{K}^n}(\tilde{x} - \alpha F(\tilde{x})) = \tilde{x}$, i.e., $R_\alpha(\tilde{x}) = 0$. Conversely, if $R_\alpha(\tilde{x}) = 0$, then $\tilde{x} = \Pi_{\mathcal{K}^n}(\tilde{x} - \alpha F(\tilde{x}))$; the characterization yields $\tilde{x} \in \mathcal{K}^n$, $-\alpha F(\tilde{x}) \in (\mathcal{K}^n)^\circ = -\mathcal{K}^n$ and $\langle -\alpha F(\tilde{x}), \tilde{x} \rangle = 0$, i.e., $F(\tilde{x}) \in \mathcal{K}^n$ and $\langle \tilde{x}, F(\tilde{x}) \rangle = 0$, which is exactly (14). $\square$

The following structural property of F and F_i is crucial for the convergence discussion: they are never strongly monotone.

Proposition 2. Let F be defined by (15) (with $F(0) = 0$), or let F_i ($i = 1, 2$) be defined by (20) (with $F_i(0) = 0$) under the standing assumptions of Lemma 7. Then

$$\langle \tilde{x}, F(\tilde{x}) \rangle = 0 \quad \text{for all } \tilde{x} \in \mathbb{R}^n \quad (24)$$

and, respectively, $\langle \tilde{x}, F_i(\tilde{x}) \rangle = 0$ for all $\tilde{x} \in \mathbb{R}^n$. In particular, the complementarity condition in (14) is automatically satisfied, so (14) reduces to the feasibility conditions $\tilde{x} \in \mathcal{K}^n$ and $F(\tilde{x}) \in \mathcal{K}^n$. Moreover, F and F_i are not strongly monotone on any set containing 0 with any modulus $\mu > 0$; in general they are not even monotone on $\mathcal{K}^n$.

Proof. For F in (15), setting $w := D\tilde{x}$, we have

$$\langle \tilde{x}, F(\tilde{x}) \rangle = \tilde{x}^T D \tilde{A} D \tilde{x} - \frac{w^T \tilde{A} w}{w^T w} \tilde{x}^T D^2 \tilde{x} = w^T \tilde{A} w - w^T \tilde{A} w = 0.$$

For F_i in (20), since $\lambda_i(\tilde{x})$ satisfies the quadratic Eq. (16), we obtain

$$\langle \tilde{x}, F_i(\tilde{x}) \rangle = \lambda_i^2 w^T \tilde{A} w + \lambda_i w^T \tilde{B} w + w^T \tilde{C} w = 0.$$

Strong monotonicity of F (resp. F_i) with modulus $\mu > 0$ would require $\langle F(\tilde{x}) - F(0), \tilde{x} - 0 \rangle \geq \mu \|\tilde{x}\|^2$ for all $\tilde{x} \neq 0$; since $F(0) = 0$ (resp. $F_i(0) = 0$) and the left-hand side equals 0 by (24), no such μ exists. Moreover, for $n = 2$, $A = \text{diag}(1, -1)$ (so that $U = D = I$ and F is given by (15)), the points $\tilde{x} = (1, 1)$ and $\tilde{y} = (1, -1)$ belong to $\mathcal{K}^2$ and a direct computation gives $\langle F(\tilde{x}) - F(\tilde{y}), \tilde{x} - \tilde{y} \rangle = -4 < 0$; hence F is not monotone on $\mathcal{K}^n$ in general. $\square$

Remark 5. The mappings in [Proposition 2](#) are positively homogeneous of degree one, i.e., $F(t\tilde{x}) = tF(\tilde{x})$ for all $t > 0$ (and similarly for F_i), and they are globally Lipschitz continuous. Indeed, writing $h(w) := \tilde{A}w - \frac{w^T \tilde{A}w}{w^T w} w$, we have $F(\tilde{x}) = Dh(D\tilde{x})$, and a direct estimate of the Jacobian in [Lemma 6](#) gives

$$\|F(\tilde{x}) - F(\tilde{y})\| \leq 6\|D\|^2 \|\tilde{A}\|_2 \|\tilde{x} - \tilde{y}\| \quad \text{for all } \tilde{x}, \tilde{y} \in \mathbb{R}^n.$$

For F_i , under the discriminant assumption [\(21\)](#), let $C_\lambda := \max_{\|w\|=1} \|\nabla \lambda_i(w)\| < \infty$ and let M_3 denote the uniform bound $|\lambda_i(w)| \leq M_3$ established in the proof of [Lemma 7](#). Estimating the Jacobian in [Lemma 8](#) yields

$$\|F_i(\tilde{x}) - F_i(\tilde{y})\| \leq \|D\|^2 (2M_3 \|\tilde{A}\|_2 + \|\tilde{B}\|_2) C_\lambda + M_3^2 \|\tilde{A}\|_2 + M_3 \|\tilde{B}\|_2 + \|\tilde{C}\|_2 \|\tilde{x} - \tilde{y}\|.$$

These Lipschitz constants do not restore the classical convergence theory, because strong monotonicity fails by [Proposition 2](#); they may nevertheless be useful for other fixed-point or error-bound analyses.

The next proposition clarifies the precise sense in which termination of [Algorithm 1](#) is meaningful: whenever the safeguarded iterates converge with vanishing fixed-reference residuals, the limit is necessarily a solution of [\(14\)](#).

Proposition 3. Let $\{\tilde{x}^k\}$ be generated by [Algorithm 1](#) with a fixed reference step $\alpha_0 > 0$, and suppose that the safeguard never reaches the warning branch, so that $\alpha_k \geq \alpha_{\min} > 0$ for all k (in [Algorithm 1](#), $\alpha_{\min} = \text{tol}_2$). If a subsequence $\{\tilde{x}^{k_j}\}$ converges to $\tilde{x}$ and $\lim_{j \rightarrow \infty} \|R_{\alpha_0}(\tilde{x}^{k_j})\| = 0$, then $\tilde{x}$ solves [\(14\)](#). In particular, whenever [Algorithm 1](#) terminates by the test $\|R_{\alpha_0}(\tilde{x}^k)\| \leq \text{tol}_1$, the output point is an approximate solution of [\(14\)](#) whose fixed-reference natural residual is at most tol_1 .

Proof. The mapping R_{α_0} is continuous on $\mathbb{R}^n$, because F and $\Pi_{\mathcal{K}^n}$ are continuous; hence $R_{\alpha_0}(\tilde{x}) = \lim_{j \rightarrow \infty} R_{\alpha_0}(\tilde{x}^{k_j}) = 0$, and [Proposition 1](#) yields that $\tilde{x}$ solves [\(14\)](#). $\square$

The iteration [\(22\)](#) is the classical projection method for the variational-inequality form of [\(14\)](#); see [\[38, Ch. 12\]](#) and [\[39\]](#). Note that [\(22\)](#) is not a gradient-projection step: it does not require the Jacobian of F , and the direction $-\alpha F(\tilde{x})$ is not the gradient of any merit function used in this paper. The explicit gradients derived in [Lemmas 6](#) and [8](#) are used in the smoothness analysis of F ; they would be needed by alternative first-order descent schemes based on the gradient of Θ_α . [Algorithm 1](#) implements [\(22\)](#) together with an adaptive step-size safeguard.

Algorithm 1 Projection method with adaptive step size for problem [\(14\)](#).

Input: Set the algorithm parameters as $\text{maxiter} = 5000$, $\text{tol}_1 = 10^{-6}$, $\text{tol}_2 = 10^{-8}$, $\alpha_0 = 1.0$, $\beta = 0.5$, set $\alpha = \alpha_0$ and choose an initial point $\tilde{x}^0$ on the slice S .

for $k = 0, 1, \dots, \text{maxiter}$ **do**

$F_k = F(\tilde{x}^k)$; $R_0 = \tilde{x}^k - \Pi_{\mathcal{K}^n}(\tilde{x}^k - \alpha_0 F_k)$;

if $\|R_0\| \leq \text{tol}_1$ **then**

return $\tilde{x}^k$

▷ fixed-reference natural-residual test, see [Proposition 1](#)

$\tilde{y} = \Pi_S(\tilde{x}^k - \alpha F_k)$;

while $\|R_{\alpha_0}(\tilde{y})\| > \|R_0\|$ **do**

▷ safeguard: nonincrease of the fixed-reference residual

$\alpha = \alpha * \beta$;

if $\alpha < \text{tol}_2$ **then**

return $\tilde{x}^k$;

▷ no certified step found; this branch was never reached in our experiments

$\tilde{y} = \Pi_S(\tilde{x}^k - \alpha F_k)$;

$\tilde{x}^{k+1} = \tilde{y}$;

Output: $\tilde{x}^{\text{maxiter}+1}$

The origin is always a fixed point of the iteration [\(22\)](#), since $F(0) = 0$; the normalization $\tilde{x}_n = 1$ excludes it, and the feasible slice S is compact. In the experiments, [Algorithm 1](#) is started from several random points on S , and the point with the smallest final fixed-reference residual is reported; the algorithm consistently produced nonzero solutions.

Regarding convergence, exactness is provided by [Proposition 1](#): $\Theta_\alpha(\tilde{x}) = 0$ if and only if $\tilde{x}$ solves [\(14\)](#). [Algorithm 1](#) terminates by the fixed-reference natural-residual test $\|R_{\alpha_0}(\tilde{x}^k)\| \leq \text{tol}_1$ with a constant step $\alpha_0 > 0$, so termination certifies an approximate solution of [\(14\)](#) in the sense of [Proposition 1](#); using a fixed reference step avoids the degeneracy that would occur if the test relied on an arbitrarily small α_k (for which R_α may vanish away from solutions). Moreover, by [Proposition 3](#), any convergent subsequence whose fixed-reference residuals tend to zero has a limit that solves [\(14\)](#). We do not claim that the safeguarded iteration decreases $\frac{1}{2}\|F(\cdot)\|^2$ in general, nor that stationarity of this quantity characterizes solutions of [\(14\)](#); such a stationarity condition is neither necessary nor sufficient for complementarity.

The classical convergence theory for projection methods, which guarantees linear convergence of the fixed-step iteration under strong monotonicity and Lipschitz continuity of F (see [\[38, Theorem 12.1.8\]](#) and [\[39\]](#)), does not apply to the present mappings. Indeed, by [Proposition 2](#), F and F_i satisfy $\langle \tilde{x}, F(\tilde{x}) \rangle = 0$ for all $\tilde{x}$, so they are never strongly monotone (and in general not even monotone), although they are globally Lipschitz with computable constants ([Remark 5](#)). [Algorithm 1](#) is therefore a heuristic: the safeguard ensures that the fixed-reference natural residual does not increase along accepted steps, and the algorithm stops as soon as it is below the tolerance. In all our experiments the safeguard terminated with certified steps (see [Table 10](#)), and the final complementarity violations are reported in [Tables 2–9](#).

Table 2

Numerical results on synthetic problems for problem (1) under the second-order cone case.

n	it	time	comp
50	122	0.028	1.0e-06
100	115	0.038	9.9e-07
150	151	0.050	9.9e-07
200	230	0.112	9.9e-07
250	202	0.166	9.9e-07
300	160	0.163	1.0e-06

For the numerical solution of problem (2) we do not use the nonlinear mapping F_i in (20) directly; instead we solve the equivalent lifted linear model of Lemma 4: find $y = (u, v) \in \mathcal{K}_{\mathcal{E}} \times \mathcal{K}_{\mathcal{E}}$ such that

$$(\lambda J - G)y \in \mathcal{K}_{\mathcal{E}}^* \times \mathcal{K}_{\mathcal{E}}^*, \quad \langle y, (\lambda J - G)y \rangle = 0,$$

where $J := \begin{bmatrix} A & 0 \\ 0 & E \end{bmatrix}$ and $G := \begin{bmatrix} -B & -C \\ E & 0 \end{bmatrix}$ with any symmetric positive definite E . Applying the transformation T of Lemma 1 blockwise, $\hat{T} := \begin{bmatrix} T & 0 \\ 0 & T \end{bmatrix}$ maps $\mathcal{K}_{\mathcal{E}} \times \mathcal{K}_{\mathcal{E}}$ onto $\mathcal{K}^n \times \mathcal{K}^n$, and the lifted model becomes the linear NSOCCP: find $\hat{y} \in \mathcal{K}^n \times \mathcal{K}^n$ such that

$$\hat{F}(\hat{y}) := \lambda \hat{J} \hat{y} - \hat{G} \hat{y} \in \mathcal{K}^n \times \mathcal{K}^n, \quad \lambda = \frac{\hat{y}^T \hat{G} \hat{y}}{\hat{y}^T \hat{J} \hat{y}},$$

with $\hat{J} := \hat{T}^T J \hat{T}$ and $\hat{G} := \hat{T}^T G \hat{T}$. By Lemma 4(ii)(a), any nonzero solution with $\lambda > 0$ yields a solution $x = (1 + \lambda)v$ of problem (2), where $v = T\hat{v}$. The mapping $\hat{F}$ has the same Rayleigh-quotient structure as F in (15), so Algorithm 1 applies directly to this linear NSOCCP of dimension $2n$; this formulation is numerically more robust than the nonlinear mapping (20).

4.2. Numerical results

All experiments were implemented in Python (NumPy) and run on a 64-bit PC with an Intel(R) Core(TM) i7-7700 CPU @ 3.60 GHz and 16.00 GB of RAM.

Synthetic test instances are generated as follows. For problem (1), A is a random $n \times n$ symmetric matrix with entries in $[-1, 1]$. For problem (2) we use the lifted linear model of the previous subsection with $A = E = I_n$ and $C = -0.5I_n$, and B is a random symmetric matrix with entries in $[-1, 1]$; this choice guarantees the hyperbolicity of the triple (A, B, C) and the strict discriminant assumption (21). The three cone subcases are the second-order cone, the circular cone with $\theta = \pi/6, \pi/3$, and the elliptic cone (see Table 1). For the elliptic cone we set $M = I_{n-1} + \delta S / \|S\|_2$ with $\delta = 0.1$ and a random symmetric $(n-1) \times (n-1)$ matrix S with entries in $[-1, 1]$, which keeps the transformation T of Lemma 1 well conditioned.

Each instance is solved by Algorithm 1 from ten random initial points on the slice S (multi-start), and the point with the smallest final fixed-reference residual is reported. Ten random seeds $1, \dots, 10$ are used; Tables 2–9 report the medians over the ten seeds (for the number of iterations and the computing time) together with the worst final violation, and Table 10 summarizes the results over all seeds. Every run is terminated by the fixed-reference natural-residual test $\|R_{a_0}(\bar{x}^k)\| \leq \text{tol}_1 = 10^{-6}$ (see Proposition 1), with $\text{maxiter} = 5000$ and safeguard tolerance $\text{tol}_2 = 10^{-8}$; the reported computing times include the overhead of the multi-start procedure.

In all tables, ‘n’, ‘it’, ‘time’ and ‘comp’ denote the problem size, the number of iterations, the total computing time in seconds and the final complementarity violation, respectively. For problem (1), ‘comp’ is the fixed-reference natural residual $\|R_{a_0}(\bar{x})\|$ evaluated at the output point; for problem (2), ‘comp’ is the violation of the complementarity conditions in (2) evaluated at the recovered solution (λ, x) of the lifted model, where $x = (1 + \lambda)v$.

Tables 2–5 report the results for problem (1) and Tables 6–9 for problem (2), for the sizes $n = 50, 100, 150, 200, 250, 300$. Table 10 reports, for each problem and each cone case, the number of successful runs out of 60 runs (10 seeds and 6 sizes), the worst final violation, the maximum number of iterations and the number of runs in which the warning branch of Algorithm 1 was reached. The method solves all instances to the required accuracy: all 240 runs for problem (1) and all 240 runs for problem (2) terminated with a violation at most 10^{-6} , and the warning branch of Algorithm 1 was never reached. The iteration counts remain of the same order of magnitude as n grows. For problem (1) the final residual is at most the tolerance, and for problem (2) the violation of the recovered solutions is typically several orders of magnitude below the tolerance (at most 2.3×10^{-9} over all runs). In particular, the method solves the nonsymmetric (circular and elliptic) cone cases with small complementarity violation.

Table 3

Numerical results on synthetic problems for problem (1) under the circular cone case with $\theta = \pi/6$.

n	it	time	comp
50	70	0.038	1.0e-06
100	70	0.013	1.0e-06
150	98	0.041	1.0e-06
200	58	0.021	1.0e-06
250	56	0.044	9.9e-07
300	48	0.024	9.6e-07

Table 4

Numerical results on synthetic problems for problem (1) under the circular cone case with $\theta = \pi/3$.

n	it	time	comp
50	269	0.100	1.0e-06
100	169	0.055	1.0e-06
150	508	0.179	1.0e-06
200	403	0.192	1.0e-06
250	464	0.303	1.0e-06
300	434	0.363	1.0e-06

Table 5

Numerical results on synthetic problems for problem (1) under the elliptic cone case.

n	it	time	comp
50	180	0.058	9.9e-07
100	117	0.032	9.8e-07
150	124	0.039	9.9e-07
200	230	0.141	1.0e-06
250	142	0.105	1.0e-06
300	166	0.152	1.0e-06

Table 6

Numerical results on synthetic problems for problem (2) under the second-order cone case.

n	it	time	comp
50	140	0.046	2.1e-14
100	138	0.103	8.4e-15
150	190	0.146	4.3e-15
200	288	0.928	3.7e-15
250	252	20.726	3.4e-15
300	206	32.821	2.8e-15

Table 7

Numerical results on synthetic problems for problem (2) under the circular cone case with $\theta = \pi/6$.

n	it	time	comp
50	74	0.037	5.5e-10
100	102	0.058	1.5e-14
150	106	0.106	3.0e-11
200	97	0.289	6.7e-15
250	98	0.630	4.5e-15
300	92	0.827	2.6e-15

Table 8

Numerical results on synthetic problems for problem (2) under the circular cone case with $\theta = \pi/3$.

n	it	time	comp
50	287	0.119	7.0e-12
100	191	2.099	1.2e-14
150	696	0.893	9.3e-15
200	458	1.765	3.6e-15
250	664	4.365	3.0e-15
300	578	9.579	2.1e-15

Table 9

Numerical results on synthetic problems for problem (2) under the elliptic cone case.

n	it	time	comp
50	236	0.060	2.2e-09
100	152	0.077	3.0e-10
150	148	0.137	7.4e-10
200	255	1.014	5.4e-10
250	192	2.613	4.0e-10
300	193	14.821	2.6e-12

Table 10

Robustness of Algorithm 1 over 10 random seeds and the sizes $n = 50, \dots, 300$ (60 runs per case). ‘success’ is the number of runs terminated with a violation at most $\text{tol}_1 = 10^{-6}$; ‘worst viol.’ is the largest final violation, ‘max it’ the largest number of iterations, and ‘ α -warn’ the number of runs in which the warning branch of Algorithm 1 was reached.

problem	cone	parameter	success	worst viol.	max it	α -warn
P1	soc		60/60	1.0e-06	416	0
P1	circular	0.5236	60/60	1.0e-06	4421	0
P1	circular	1.0472	60/60	1.0e-06	1345	0
P1	elliptic		60/60	1.0e-06	3590	0
P2	soc		60/60	2.1e-14	858	0
P2	circular	0.5236	60/60	5.5e-10	165	0
P2	circular	1.0472	60/60	7.0e-12	1698	0
P2	elliptic		60/60	2.2e-09	692	0

5. Concluding remarks

In this paper, we focused on solvability conditions and model reformulations for two types of ellipsoidal cone eigenvalue complementarity problems, which can be viewed as natural generalizations of second-order cone eigenvalue complementarity problems. Their regularized model reformulations are expressed as NSOCCPs by leveraging connections between ellipsoidal cones and second-order cones, and are solved by a customized projection method with an exact residual-based stopping rule. For problem (2), the numerical solution is obtained from an equivalent lifted linear model over the product of second-order cones (Lemma 4), which is numerically more robust than the direct nonlinear reformulation. Numerical experiments with multiple random starts and seeds show that the method terminates with complementarity violations below the tolerance on all tested instances. Establishing verifiable conditions under which the safeguarded projection iteration converges (e.g., suitable error-bound or descent properties of the natural-residual merit function) is an interesting direction for future research. Future work also includes extending the approach to other nonsymmetric cones and addressing the large-scale dense matrix issue via structure-exploiting transformations.

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Data availability

No data was used for the research described in the article.

References

- [1] Y. Lu, J.S. Chen, The variational geometry, projection expression and decomposition associated with ellipsoidal cones, *J. Nonlinear Convex Anal.* 20 (2019) 715–738.
- [2] M.S. Lobo, L. Vandenbergh, S. Boyd, H. Lebre, Applications of second-order cone programming, *Linear Algebra Appl.*, 284 (1998) 193–228. [https://doi.org/10.1016/S0024-3795\(98\)10032-0](https://doi.org/10.1016/S0024-3795(98)10032-0)
- [3] M. Fukushima, Z.Q. Luo, P. Tseng, Smoothing functions for second-order-cone complementarity problems, *SIAM J. Optim.* 12 (2002) 436–460. <https://doi.org/10.1137/S105262340038036>
- [4] F. Alizadeh, D. Goldfarb, Second-order cone programming, *Math. Program.* 95 (2003) 3–51. <https://doi.org/10.1007/s10107-002-0339-5>
- [5] J.S. Chen, X. Chen, P. Tseng, Analysis of nonsmooth vector-valued functions associated with second-order cone, *Math. Program.*, 101 (2004) 95–117. <https://doi.org/10.1007/s10107-004-0538-3>
- [6] J.S. Chen, P. Tseng, An unconstrained smooth minimization reformulation of second-order cone complementarity problem, *Math. Program.* 104 (2005) 293–327. <https://doi.org/10.1007/s10107-005-0617-0>
- [7] J.F. Bonnans, H.C. Ramirez, Perturbation analysis of second-order cone programming problems, *Math. Program.* 104 (2005) 205–227. <https://doi.org/10.1007/s10107-005-0613-4>
- [8] Y.L. Chang, C.Y. Yang, J.S. Chen, Smooth and nonsmooth analysis of vector-valued functions associated with circular cones, *Nonlinear Anal.* 85 (2013) 160–173. <https://doi.org/10.1016/j.na.2013.01.017>
- [9] J.C. Zhou, J.S. Chen, Properties of circular cone and spectral factorization associated with circular cone, *J. Nonlinear Convex Anal.* 14 (2013) 807–816.
- [10] J.C. Zhou, Y.L. Chang, J.S. Chen, The H-differentiability and calmness of circular cone functions, *J. Global Optim.* 63 (2015) 811–833. <https://doi.org/10.1007/s10898-015-0312-5>
- [11] J.C. Zhou, J.S. Chen, Monotonicity and circular cone monotonicity associated with circular cones, *Set-Valued Var. Anal.* 25 (2017) 211–232. <https://doi.org/10.1007/s11228-016-0374-7>
- [12] J.C. Zhou, J.Y. Tang, J.S. Chen, Parabolic second-order directional differentiability in the Hadamard sense of the vector-valued functions associated with circular cones, *J. Optim. Theory Appl.* 172 (2017) 802–823. <https://doi.org/10.1007/s10957-016-0935-9>
- [13] Y. Lu, J.S. Chen, N. Zhang, No gap second-order optimality conditions for circular conic programs, *Numer. Funct. Anal. Optim.* 40 (2019) 1113–1135. <https://doi.org/10.1080/01630563.2018.1552965>
- [14] Y. Lu, H.M. Ma, D.Y. Xue, J.S. Chen, Augmented Lagrangian method for nonlinear circular conic programs: a local convergence analysis, *Optimization* (2024). <https://doi.org/10.1080/02331934.2024.2370426>
- [15] B. Alzalg, M. Pirhaji, Elliptic cone optimization and primal-dual path-following algorithms, *Optimization* 66 (2017) 2245–2274. <https://doi.org/10.1080/02331934.2017.1360888>
- [16] R.J. Stern, H. Wolkowicz, Invariant ellipsoidal cones, *Linear Algebra Appl.* 150 (1991) 81–106. [https://doi.org/10.1016/0024-3795\(91\)90161-0](https://doi.org/10.1016/0024-3795(91)90161-0)
- [17] Y. Lu, J.S. Chen, Smooth analysis on cone function associated with ellipsoidal cone, *J. Nonlinear Convex Anal.* 21 (2020) 1327–1347.
- [18] X.H. Miao, Y. Lu, J.S. Chen, Construction of merit functions for ellipsoidal cone complementarity problem, *Pac. J. Optim.* 16 (2020) 547–565.
- [19] R.J. Stern, H. Wolkowicz, Exponential nonnegativity on the ice cream cone, *SIAM J. Matrix Anal. Appl.* 12 (1991) 160–165. <https://doi.org/10.1137/0612012>
- [20] R. Bhattacharya, J. Fung, A. Tiwari, Ellipsoidal cones and rendezvous of multiple agents, in: *Proceedings of the 43rd IEEE Conference on Decision and Control*, 2004, pp. 171–176. <https://doi.org/10.1109/CDC.2004.1428625>
- [21] A. Seeger, J. Vidal-Nuñez, Measuring centrality and dispersion in directional datasets: the ellipsoidal cone covering approach, *J. Global Optim.* 68 (2017) 279–306. <https://doi.org/10.1007/s10898-016-0464-y>
- [22] F. Chatelin, *Eigenvalues of Matrices*, John Wiley & Sons, New York, 1993. <https://doi.org/10.1137/1.9781611972467>
- [23] J.A.C. Martins, S. Barbarin, M. Raous, A. Pinto da Costa, Dynamic stability of finite dimensional linearly elastic systems with unilateral contact and coulomb friction, *Comput. Methods Appl. Mech. Eng.* 177 (1999) 289–328. [https://doi.org/10.1016/S0045-7825\(98\)00386-7](https://doi.org/10.1016/S0045-7825(98)00386-7)
- [24] J.A.C. Martins, A. Pinto da Costa, Stability of finite-dimensional nonlinear elastic systems with unilateral contact and friction, *Int. J. Solids Struct.* 37 (2000) 2519–2564. [https://doi.org/10.1016/S0020-7683\(98\)00291-1](https://doi.org/10.1016/S0020-7683(98)00291-1)
- [25] I.N. Figueiredo, A complementarity eigenproblem in the stability analysis of finite dimensional elastic systems with frictional contact, in: M.C. Ferris, O.L. Mangasarian, J.-S. Pang (Eds.), *Complementarity: Applications, Algorithms and Extensions*, Kluwer Academic Publishers, Dordrecht, 2001. <https://doi.org/10.1007/978-1-4757-3279-5>
- [26] J.A.C. Martins, A. Pinto da Costa, I.N. Figueiredo, J.J. Judice, The directional instability problem in systems with frictional contacts, *Comput. Methods Appl. Mech. Eng.* 193 (2004) 357–384. <https://doi.org/10.1016/j.cma.2003.09.013>
- [27] M. Queiroz, J.J. Judice, J.C. Humes, The symmetric eigenvalue complementarity problem, *Math. Comput.* 73 (2004) 1849–1863. <https://doi.org/10.1090/S0025-5718-03-01614-4>
- [28] J.J. Judice, H.D. Serali, I.M. Ribeiro, The eigenvalue complementarity problem, *Comput. Optim. Appl.* 37 (2007) 139–156. <https://doi.org/10.1007/s10589-007-9017-0>
- [29] A. Pinto da Costa, A. Seeger, Cone-constrained eigenvalue problems: theory and algorithms, *Comput. Optim. Appl.* 45 (2010) 25–57. <https://doi.org/10.1007/s10589-008-9167-8>
- [30] A. Seeger, Quadratic eigenvalue problems under conic constraints, *SIAM J. Matrix Anal. Appl.* 32 (2011) 700–721. <https://doi.org/10.1137/100801780>
- [31] C.P. Bras, A.N. Iusem, J.J. Judice, On the quadratic eigenvalue complementarity problem, *J. Global Optim.* 66 (2016) 153–171. <https://doi.org/10.1007/s10898-014-0260-5>
- [32] L.M. Fernandes, M. Fukushima, J.J. Judice, H.D. Serali, The second-order cone eigenvalue complementarity problem, *Optim. Methods Softw.* 31 (2016) 24–52. <https://doi.org/10.1080/10556788.2015.1040156>
- [33] J.J. Judice, M. Fukushima, A. Iusem, J.M. Martinez, V. Sessa, An alternating direction method of multipliers for the eigenvalue complementarity problem, *Optim. Methods Softw.* 36 (2020) 337–370. <https://doi.org/10.1080/10556788.2020.1734804>
- [34] C. Bras, M. Fukushima, A. Iusem, J.J. Judice, On the quadratic eigenvalue complementarity problem over a general convex cone, *Appl. Math. Comput.* 271 (2015) 391–403. <https://doi.org/10.1016/j.amc.2015.09.014>
- [35] A. Seeger, Eigenvalue analysis of equilibrium processes defined by linear complementarity conditions, *Linear Algebra Appl.* 292 (1999) 1–14. [https://doi.org/10.1016/S0024-3795\(99\)00004-X](https://doi.org/10.1016/S0024-3795(99)00004-X)
- [36] A. Seeger, M. Torki, On eigenvalues induced by a cone constraint, *Linear Algebra Appl.* 372 (2003) 181–206. [https://doi.org/10.1016/S0024-3795\(03\)00553-6](https://doi.org/10.1016/S0024-3795(03)00553-6)
- [37] J.P. Aubin, H. Frankowska, *Set-Valued Analysis*, Birkhäuser, Boston, 2003. <https://doi.org/10.1007/978-0-8176-4848-0>
- [38] F. Facchinei, J.-S. Pang, *Finite-Dimensional Variational Inequalities and Complementarity Problems*, vol. I and II, Springer, New York, 2003. <https://doi.org/10.1007/b97543>
- [39] D.P. Bertsekas, J.N. Tsitsiklis, *Parallel and Distributed Computation: Numerical Methods*, Prentice-Hall, Englewood Cliffs, NJ, 1989.